

GENESIS: A Hadean Earth Titan-Mass Impact Model — A Quantitative Hypothesis for Earth’s Volatile Budget, Atmospheric Reset, and Post-Impact Prebiotic Chemistry

An Inverse-Calibrated Framework, an Exploratory Technical Companion and Working Paper — Expanded Revision

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Code and data:

<https://github.com/ChristosTsirkas/genesis-earth-hadean-titan-impact-model>

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Accompanying material

Theatrical play libretto:

<https://github.com/ChristosTsirkas/genesis-earth-hadean-titan-impact-model/blob/main/docs/GENESIS%20%E2%80%94%20Complete%20Theatrical%20Play%20%E2%80%94%20Musical%20Score%20%E2%80%94%20Composer%20Dossier.pdf>

Additional, computer-generated support files for the music of this project:

<https://github.com/ChristosTsirkas/genesis-earth-hadean-titan-impact-model/tree/main/music>

A rendered realization of the “*GENESIS*” musical score is also available online, allowing the musical component to be heard independently of a staged performance.

<https://soundcloud.com/chris-t-331652374>

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Lekkas, Demetrios E. "*I mathematiki theoria tis mousikis: Pythagoreio kai fysiko systima*" [*The mathematical theory of music: Pythagorean and natural system*]. PhD diss., National and Kapodistrian University of Athens, 1995. National Archive of PhD Theses.

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Status of the model

The collision described here is a deliberately testable working hypothesis, not an established event in terrestrial chronology. The paper asks whether a Titan-mass, volatile-rich body could in principle generate a physically coherent sequence of volatile delivery, atmospheric loss/reconstitution, ocean condensation, and prebiotic chemical opportunity. Wherever the current literature does not constrain a parameter, the analysis exposes that parameter explicitly rather than treating it as measured.

Abstract

Earth's Hadean volatile inventory and the origin of its nitrogen-dominated atmosphere remain coupled problems in planetary accretion, atmospheric escape, isotope geochemistry, and prebiotic chemistry. This paper develops a deliberately non-canonical end-member hypothesis: a late giant impact at approximately 4.1 Ga involving a Titan-mass ($M_i = 1.345 \times 10^{23}$ kg) volatile-rich icy body. The revised calculation treats the event as a giant impact rather than an enlarged crater-forming impact. At a reference velocity of 15 km s⁻¹, the impact kinetic energy is 1.51×10^{31} J, approximately 6.8% of Earth's gravitational binding energy and many orders of magnitude above Chicxulub-class events. A nominal 50 wt.% water inventory corresponds to 6.73×10^{22} kg, or ~48 modern terrestrial ocean masses; consequently, the scientifically relevant quantity is not gross delivery but the retained fraction after immediate ejecta loss, silicate-volatile partitioning, photolysis, and hydrodynamic escape. Likewise, an assumed ammonia-bearing fraction cannot be converted directly into the present nitrogen atmosphere: even modest NH₃ inventories can generate hundreds of bars of N₂ if fully retained, requiring strong subsequent loss and/or sequestration.

The model is therefore reformulated as a coupled mass-balance problem, supported by a computational verification tier and an executable numerical pipeline. We derive numerical constraints on impact energetics, volatile retention, equivalent surface pressure, nitrogen yield, radiative cooling, and isotope mixing. Runs 7 and 8 introduce high-precision N-body infrastructure using REBOUND (IAS15/MERCURIUS) to establish structural transport baselines, while Run 9 expands the environment into a literature-anchored multi-planet instability framework. Concurrently, a standalone preparation utility establishes a 96-case Cartesian matrix for future cluster-scale SWIFT/SPH hydrodynamic campaigns to evaluate high-obliquity volatile stripping and selective noble-gas depletion. The D/H problem remains an open mixing constraint: Enceladus water is measurably D-rich relative to terrestrial oceans, while Titan's directly measured D/H values primarily sample methane or hydrogen rather than its deep-water reservoir. The paper also separates prebiotic chemical plausibility from biological inference; robust evidence for life is substantially younger than 4.1 Ga, while older isotopic claims remain disputed.

The result is a falsifiable framework rather than a narrative proof. Its strongest prediction is not that a specific 'Titan Twin' is known to have struck Earth, but that any viable version of such a late volatile-rich giant impact must occupy a narrow joint parameter space in impact velocity and angle, water and ammonia abundance, volatile retention efficiency, atmospheric escape efficiency, and post-impact redox state. Those constraints can be tested against Hadean zircon geochemistry, mantle noble gases and nitrogen isotopes, lunar chronology, dynamical simulations of giant-planet instability, and high-pressure shock experiments.

Keywords: *Hadean Earth; giant impact; volatile delivery; Titan; icy satellites; D/H; nitrogen atmosphere; ammonia; magma ocean; atmospheric escape; origin of life*

Scope and Structure of This Paper

This paper examines a specific end-member hypothesis: whether a late, high-obliquity impact by a Titan-mass volatile-rich body could be made consistent with Earth's present volatile inventory, atmosphere, isotopic constraints, and early chemical environment. It does not claim that such an event has been established in Earth's history.

The analysis proceeds in stages. It first defines the physical and geochemical constraints that any viable impact scenario must satisfy. It then evaluates the associated energy, volatile retention, atmospheric loss, nitrogen production, cooling, isotope mixing, and prebiotic chemical consequences. Appendix D records the deterministic numerical results and explains how each result can be reproduced.

The present computational work includes the reduced-order calculations and tested N-body transport infrastructure. The production-scale planetary-instability ensembles and 96-case SWIFT/SPH hydrodynamic campaign require high-performance computing and remain proposed tests rather than completed simulations. This distinction is maintained throughout the paper so that calculated results, numerical preparations, and future validation work are not presented as equivalent forms of evidence.

Because this framework also functions as the quantitative scientific companion to a dramatic and musical work (Tsirkas, 2026), it deliberately investigates an end-member evolutionary pathway. Such scenarios explore limiting behaviour rather than establish the most statistically probable history. The computational pipeline does not demonstrate that this specific Titan-mass impact occurred; it identifies the severe, non-linear constraints and fine-tuning that the proposed scenario would have to satisfy to remain compatible with the modern Earth.

AI-Assisted Research, Computational Development, and Author Disclosure

This paper was developed as an independent interdisciplinary research project with substantial assistance from generative artificial-intelligence systems. The author's professional background is in software development rather than in astrobiology, planetary science, geochemistry, celestial mechanics, or impact hydrodynamics. This distinction is stated explicitly, because the present work addresses a technically demanding problem across several scientific disciplines and should not be read as a claim of formal specialist qualification in each of them.

The central hypothesis, its overall physical scenario, the questions investigated, the selection of modelling objectives, the interpretation pursued, and the decision to develop and publish the project belong to the author. Artificial-intelligence systems were used extensively as research and development tools: to map relevant scientific literature; explain and connect concepts from different disciplines; assist in the formulation and dimensional checking of equations; develop, debug, and revise the accompanying Python code; organize parameter studies and calibration runs; examine numerical outputs for internal consistency; identify weaknesses and possible counterarguments; and assist with the structure, technical language, and successive revision of the manuscript.

AI-generated material was not accepted automatically. Proposed equations, assumptions, citations, parameter values, computational procedures, interpretations, and textual revisions were selected, modified, rejected, or retained through an iterative process directed by the author. Nevertheless, such review must not be confused with independent validation by specialists in the relevant scientific fields. The calculations and reduced-order simulations reported here constitute an internally testable exploratory framework, not a substitute for peer review, laboratory evidence, expert replication, or high-resolution hydrodynamic modelling.

Artificial-intelligence systems are not presented as authors, scientific authorities, or sources of empirical evidence. Responsibility for the decision to present the hypothesis, for the organization of the argument, for the disclosed assumptions and limitations, and for the final contents of the paper remains with the named human author. To facilitate external assessment, the computational procedures, parameter choices, calibration history, generated outputs, and principal limitations are documented alongside the manuscript wherever practicable.

Accompanying material

This paper forms part of the wider “*GENESIS*” project, which extends the scientific hypothesis and computational framework developed here into a separate dramatic and musical work. The companion volume presents the same underlying conception through theatre and music, while remaining methodologically independent from the scientific argument of the present paper.

1. Theatrical Play, Musical Score, Composer Dossier, and Presentation

The artistic counterpart of the project is distributed separately as “*GENESIS — Complete Theatrical Play — Musical Score — Composer Dossier*”. It contains the complete 544-verse theatrical play libretto in ancient attic and in English, its synchronized musical score, the Composer Dossier documenting the work’s compositional and structural system, and the accompanying project presentation. Together, these materials constitute the dramatic, musical, and presentational realization of the scientific conception developed in this paper.

Download or read the complete companion volume at the following URL:

<https://github.com/ChristosTsirkas/genesis-earth-hadean-titan-impact-model/tree/main/docs/GENESIS — Complete Theatrical Play — Musical Score — Composer Dossier.pdf>

2. **The computational framework** accompanying the paper provides the code, configuration, calibration history, and numerical outputs used for its reduced-order local screening calculations. It includes the computational stages for delivery screening, impact and volatile partitioning, thermal evolution, atmospheric escape and isotope evolution, noble-gas retention, and post-impact chemistry, together with the archived results of the successive calibration and reproducibility runs discussed in the paper.

View/download the GENESIS computational code at the following GitHub repository:

<https://github.com/ChristosTsirkas/genesis-earth-hadean-titan-impact-model>

The code repository is provided both for transparency and for reproducibility. Readers may inspect the implementation, numerical assumptions, calibrated surrogate parameters, intermediate outputs, and final result matrices rather than relying solely upon the values reported in the manuscript.

3. A rendered realization of the “GENESIS” music score is also available online, allowing the musical component to be heard independently of a staged performance.

Listen to “GENESIS” on SoundCloud:

<https://soundcloud.com/chris-t-331652374>

This scientific paper and computational material constitute the authoritative record of the project’s scientific claims, assumptions, calculations, and limitations. The play and musical score constitute their independent artistic transformation and are not themselves offered as scientific evidence.

Together, the separately distributed components of “GENESIS” comprise:

- *Scientific Paper* — the hypothesis, quantitative analysis, results, limitations, and falsifiable predictions, distributed in this document.
- *Computational Framework* — the executable models, calibration history, reproducibility material, and numerical results.
- *Theatrical Play / Libretto* — the dramatic and mythological transformation of the scientific conception.
- *Musical Score* — the musical realization of the dramatic work, distributed together with the theatrical play.
- *Composer Dossier* — the complete technical and structural documentation of the 544-verse score, distributed together with the theatrical play.
- *The Music* — the rendered realization of the score, available for online listening on SoundCloud.

Each component may be read, examined, performed, or evaluated independently; taken together, they constitute the complete “GENESIS” project.

1. Scope, Terminology, and Corrections to the First Draft

The first draft framed the event as a Late Heavy Bombardment impact and used several single-number estimates as though they were observationally constrained. The present revision keeps the underlying hypothesis but changes its scientific status. First, the classical concept of a narrow 3.9 Ga lunar cataclysm is debated; modern chronology permits intense early bombardment without requiring a unique spike. Accordingly, ‘late Hadean bombardment’ is preferred below, while ‘LHB’ is retained only when discussing the historical literature [1–3]. Second, a Titan-mass impactor is a planetary giant-impact regime. Its global consequences must be evaluated with hydrodynamic impact physics, not crater-energy analogy [4,5]. Third, neither the primordial D/H of Titan’s deep water nor a 5 wt.% bulk ammonia content is measured. Those quantities are therefore promoted to free parameters, constrained by analogues and models rather than asserted as facts [6–10].

This distinction is not cosmetic. A useful hypothesis paper should make it easy for a critic to identify what would falsify it. The central question is therefore conditional: if a volatile-rich body with mass of order 10^{23} kg reached Earth at ~ 4.1 Ga, is there a region of parameter space in which the retained water, nitrogen, redox state, and cooling history are simultaneously consistent with later terrestrial constraints? Sections 2–8 answer that question analytically at zeroth order; Section 9 defines the simulations needed to move beyond it.

1.1 Reference case and deliberately free parameters

Parameter	Symbol	Reference case	Status
Impactor mass	M_i	1.345×10^{23} kg	Titan mass; fixed scale
Earth mass	M_E	5.972×10^{24} kg	Known
Impact speed	v_i	15 km s ⁻¹	Free; plausible giant-impact scale
Impact angle	θ	Oblique/grazing family	Preferred Moon-survival regime; exact θ/b free and simulation-dependent
Water mass fraction	x_{H_2O}	0.50	End-member assumption
Ammonia mass fraction	x_{NH_3}	0.005–0.05	Broad exploratory interval
Immediate retained water fraction	η_{H_2O}	0.01–0.50	Unknown; simulation-dependent
NH ₃ → N ₂ conversion fraction	χ_N	0–1	Unknown; chemistry-dependent
Post-impact N ₂ retention fraction	η_{N_2}	10 ⁻⁴ –0.5	Unknown; escape/sequestration
Impactor water D/H	R_i	free	Not measured for Titan deep water

Parameter	Symbol	Reference case	Status
Impact epoch	t _i	~4.1 Ga	Hypothesis, not established event

2. Giant-Impact Dynamics and Energetic Scale

For the reference case, $M_i/M_E = 0.0225$. This is far above the mass ratios of ordinary asteroid impacts and enters the regime in which the target's global deformation, vapor plume, atmosphere, and mantle response become coupled. The first-order kinetic energy is

$$E_k = \frac{1}{2} M_i v_i^2 = 1.51 \times 10^{31} \text{ J (for } M_i = 1.345 \times 10^{23} \text{ kg and } v_i = 15 \text{ km s}^{-1}\text{)}.$$

For comparison, the gravitational binding energy of a uniform-density Earth is approximately

$$U_E \approx (3/5) G M_E^2/R_E = 2.24 \times 10^{32} \text{ J},$$

so $E_k/U_E \approx 0.0675$. The exact fraction of impact energy coupled to mantle melting, vaporization, ejecta, rotation, and escaping material depends strongly on geometry and equations of state, but the scale alone excludes language such as 'a hundred Chicxulubs.' A Titan-mass collision is roughly 10^7 – 10^8 times a $\sim 10^{23}$ – 10^{24} J Chicxulub-class event. It is therefore reasonable to expect extensive or global melting and a massive silicate-volatile vapor atmosphere, although whether the mantle is globally homogenized requires numerical calculation [4,5,11].

2.1 Mutual escape velocity and admissible impact speeds

An impact speed should be decomposed into gravitational focusing and hyperbolic encounter speed. If the impactor radius is approximated by Titan's radius $R_i \approx 2.575 \times 10^6$ m, the mutual escape speed at contact is

$$v_{\text{esc,mut}} = [2G(M_E + M_i)/(R_E + R_i)]^{1/2} \approx 9.55 \text{ km s}^{-1}.$$

Thus a 15 km s⁻¹ collision corresponds to a substantial but not extreme excess over mutual escape. The implied encounter speed at infinity is $v_\infty \approx (v_i^2 - v_{\text{esc,mut}}^2)^{1/2} \approx 11.6 \text{ km s}^{-1}$. A dynamical origin in the giant-planet region must therefore do more than merely eject a satellite: it must deliver an Earth-crossing orbit with the correct energy and phase. No published dynamical study presently demonstrates the specific path assumed here. That absence is a major uncertainty, not a minor missing citation (Section 9.1).

2.2 Dynamical source reservoirs: giant-planet instability as context

Two source classes provide physically motivated starting conditions for a dedicated delivery calculation. The first is a large icy planetesimal or planetary embryo originating beyond the giant planets. Nice-model-type instabilities demonstrably scatter trans-Neptunian material inward and can implant outer-system objects into the asteroid region [36,37]. This establishes a real transport mechanism from the trans-Neptunian reservoir toward the inner Solar System, although it does not by itself demonstrate a Titan-mass Earth impact. The second, more speculative class is a primordial satellite or circumplanetary embryo destabilized during close encounters among giant planets. Numerical studies show that giant-planet encounters can strongly excite or disrupt regular-satellite systems [38], so satellite destabilization is dynamically conceivable. No published calculation currently follows a Titan-mass escaped satellite from a giant planet to a ~ 4.1 Ga Earth collision, and this distinction must remain explicit.

Chronology is an additional constraint. The original Nice model was historically associated with a late instability, but several modern dynamical models instead favor an instability triggered during or shortly after protoplanetary-gas dispersal, only a few to ~ 10 Myr after Solar-System formation [36]. Such early-instability solutions do not naturally generate a unique 4.1 Ga source event. Therefore this paper uses "giant-planet instability" as a dynamical mechanism, not as chronological evidence for the proposed impact epoch. A viable N-body model must either identify a long-lived metastable reservoir capable of

releasing the body hundreds of Myr later, or demonstrate another late scattering pathway consistent with current bombardment chronology.

2.3 Angular momentum is a first-order constraint

A late impact this large cannot be evaluated only by energy. An oblique impact carries angular momentum $L_i \approx M_i v_i b$, where b is the impact parameter. Because the Moon is already present in the proposed ~ 4.1 Ga chronology, impact geometry becomes a first-order survival constraint rather than a secondary nuisance parameter. The admissible family should preferentially occupy oblique to grazing geometries that reduce deep coupling and avoid approximately symmetric core-to-core coupling, while still satisfying volatile-retention requirements. This preference is a hypothesis to be tested, not a demonstrated guarantee of lunar survival. Any viable solution must reproduce acceptable terrestrial spin, Earth–Moon angular momentum, and post-impact lunar orbital evolution with N-body + SPH calculations.

2.4 Moon-survival-constrained grazing impact regime

The revised reference architecture therefore replaces a single nominal 45° impact with a Moon-survival-constrained oblique/grazing family. The physical motivation is not that grazing geometry automatically preserves the Moon, but that geometry controls how a Titan-mass projectile partitions its momentum and energy among deep target deformation, tangential shear, escaping ejecta, vapor plume, projectile survival, atmosphere/ocean coupling, and reaccretion. At fixed M_i and v_i , changing θ and b can therefore change the outcome qualitatively, including transitions among accretionary, erosive, and hit-and-run-like regimes.

This geometrical revision is especially important because the hypothesis is an over-delivery problem. A strongly oblique encounter can, in principle, allow a larger fraction of the icy projectile and shocked surface material to enter asymmetric ejecta/vapor trajectories rather than being incorporated into a fully mixed deep reservoir. It may also reduce symmetric core-to-core coupling and global homogenization relative to a more symmetric, lower-obliquity collision of the same kinetic energy. These are expected tendencies, not quantified results for the Titan-Twin case; their magnitude must be obtained from the SPH suite in Section 9.2.

Accordingly, the key dynamical variables are now treated jointly as $(v_i/v_{\text{esc}}, \theta, b/R_E)$ rather than reducing geometry to one reference angle. A viable solution must simultaneously satisfy volatile partitioning, terrestrial spin, total Earth–Moon angular momentum, lunar semimajor-axis/eccentricity/inclination evolution, and survival of the existing Moon. The grazing family is therefore a constrained search region of the hypothesis, not an imposed single-angle solution.

3. Volatile Delivery as a Retention Problem

The nominal volatile inventory of a Titan-like object is so large that the paper's central problem reverses. The question is not whether enough water can be supplied; it is whether little enough survives to remain compatible with Earth. If $x_{\text{H}_2\text{O}}$ denotes the impactor water mass fraction, the gross delivered mass is

$$M_{\text{H}_2\text{O},0} = x_{\text{H}_2\text{O}} M_i.$$

At $x_{\text{H}_2\text{O}} = 0.50$, $M_{\text{H}_2\text{O},0} = 6.73 \times 10^{22}$ kg, about 48 modern ocean masses (using $M_{\text{ocean}} \approx 1.4 \times 10^{21}$ kg). The post-impact surface inventory is instead

$$M_{\text{H}_2\text{O},\text{ret}} = \eta_{\text{H}_2\text{O}} x_{\text{H}_2\text{O}} M_i,$$

where $\eta_{\text{H}_2\text{O}}$ is a compound retention factor that implicitly includes immediate ejecta, impact-vapor escape, dissolution into melt, later mantle storage, photolysis, hydrogen escape, and reaccrretion. To add only one present ocean from the impactor at $x_{\text{H}_2\text{O}} = 0.50$ requires $\eta_{\text{H}_2\text{O}} \approx 0.0208$. Five oceans require $\eta_{\text{H}_2\text{O}} \approx 0.104$. Thus even a few-percent retention efficiency is geochemically important.

For the grazing-impact revision, $\eta_{\text{H}_2\text{O}}$ remains useful as a zeroth-order scalar, but the numerical model should resolve the water budget into a partition vector rather than treating all non-surface water as one sink:

$$\vec{\eta}_w = (\eta_{\text{escape}}, \eta_{\text{vapor/plume}}, \eta_{\text{mantle}}, \eta_{\text{surface}}, \eta_{\text{reaccrrete}}).$$

The components are geometry-dependent and need not evolve independently. In particular, an oblique/grazing impact can place substantial volatile mass into asymmetric plume/ejecta trajectories while leaving a different fraction available for mantle dissolution, atmospheric processing, later condensation, or reaccrretion. This partition is one of the principal outputs required from Section 9.2.

3.1 Equivalent pressure exposes the scale

If the nominal water mass were temporarily distributed as a hydrostatic atmosphere, its equivalent surface pressure would be

$$P_{\text{eq}} = M g / (4\pi R E^2).$$

For 6.73×10^{22} kg of H_2O , $P_{\text{eq}} \approx 1.29 \times 10^9$ Pa $\approx$ 12,900 bar. This does not mean a stable 12,900-bar steam atmosphere necessarily forms: much of the material would be in supercritical fluid, silicate vapor, melt, ejecta, or escaping flow. It does show why a one-layer 'steam blanket' calculation is inadequate. The coupled silicate-volatile atmosphere can remain hot for 10^3 – 10^6 + yr depending on inventory, opacity, albedo, melt convection, and escape [12–14].

3.2 Lower-bound cooling estimate and why the old 1.1 Myr value was misleading

The first draft divided the latent heat of water by a 280 W m^{-2} radiation limit and obtained ~ 1.1 Myr. Re-evaluating that expression with the stated quantities gives a much shorter latent-heat-only timescale, about 3.4×10^3 yr:

$$\tau_{\text{lat}} \approx M_{\text{H}_2\text{O}} L_v / (4\pi R E^2 F_{\text{OLR}}) \approx 1.06 \times 10^{11} \text{ s} \approx 3.4 \text{ kyr}.$$

That number is not a realistic condensation time. It is a deliberately incomplete lower bound because it neglects the vastly larger sensible and latent heat of the silicate mantle, impact vapor, changing outgoing-longwave-radiation (OLR), absorbed sunlight, dissolution/exsolution, and continuing internal heat flux. Modern magma-ocean models obtain lifetimes from thousands to millions of years depending on atmospheric mass and radiative structure [12–14]. The correct role of the $\sim 280 \text{ W m}^{-2}$ Simpson–

Nakajima-type limit is therefore as one boundary condition in a coupled energy balance, not as a standalone stopwatch.

4. Hydrogen Isotopes: From “Resolution” to a Mixing Constraint

The isotopic section requires the strongest conceptual revision. Terrestrial ocean water has $D/H \approx 1.558 \times 10^{-4}$. Some carbonaceous chondrites and at least one Jupiter-family comet are close to this value [15]. Enceladus plume water, however, was measured at $D/H \approx 2.9(+1.5/-0.7) \times 10^{-4}$, close to cometary rather than terrestrial water [6]. Titan’s measured D/H values derive mainly from methane or molecular hydrogen and are of order $1.0\text{--}1.6 \times 10^{-4}$ [7,8]; they cannot simply be equated to the D/H of Titan’s deep water reservoir. Consequently, the claim that a hypothetical 4.1 Ga Titan analogue ‘perfectly matches SMOW’ is currently unsupported.

4.1 Two-reservoir isotope mass balance

Let R_{E0} be the D/H ratio of Earth’s retained water immediately before the impact, R_i that of retained impactor water, and f the fraction of the final exchangeable surface-water hydrogen inventory contributed by the impactor after all preferential loss. To first order,

$$R_f \approx (1 - f)R_{E0} + f R_i,$$

provided the reservoirs are well mixed and hydrogen concentrations are accounted for consistently. Rearrangement gives

$$f \approx (R_f - R_{E0})/(R_i - R_{E0}).$$

This equation makes the hypothesis testable. If R_i were Enceladus-like and substantially above SMOW, a large surviving impactor fraction would tend to raise Earth’s oceanic D/H unless countered by a lower- D/H pre-impact reservoir, isotope-selective escape, mantle exchange, or later delivery. Conversely, if R_i were genuinely SMOW-like, the isotope constraint weakens. The required calculation is therefore a coupled isotope/escape model, not a backward Rayleigh extrapolation from Titan’s modern methane.

4.2 Rayleigh fractionation: useful, but only for a defined reservoir

Rayleigh evolution can be written as $R = R_0 f_r^{(\alpha-1)}$, where f_r is the remaining fraction of a specified reservoir and α is the isotope fractionation factor between escaping and retained material. The equation is valid only when the reservoir, escape flux, and fractionation process are physically defined. In the first draft, $f \approx 0.95$ at 4.1 Ga was assumed without a coupled atmospheric-history model. That assumption is removed. A future calculation should track at least H_2O , H_2 , CH_4 , HDO/HD , upper-atmosphere temperature, EUV flux, and exchange with the magma ocean. Section 9.3 specifies the minimum implementation.

5. Nitrogen and Ammonia: Stoichiometry, Pressure, and Redox

Ammonia is chemically interesting because it is both a reduced nitrogen carrier and an antifreeze in water-rich planetary materials. It is not, however, established that Titan contains 5 wt.% NH₃ in bulk. Models span a wide range: older condensation models permitted large ammonia-hydrate fractions, Enceladus plume measurements show ~sub-percent ammonia relative to water, and Titan ocean models have explored values from a few percent to >10 wt.% under different assumptions [6,9,10]. The revised paper therefore explores $x_{\text{NH}_3} = 0.005\text{--}0.05$ as an end-member interval rather than assigning a measured 5% value.

5.1 Maximum molecular-nitrogen yield

If net high-temperature chemistry is represented schematically by $2\text{NH}_3 \rightarrow \text{N}_2 + 3\text{H}_2$, the maximum N₂ mass generated from an ammonia mass M_{NH_3} is

$$M_{\text{N}_2, \text{max}} = (28/34) M_{\text{NH}_3} = 0.8235 M_{\text{NH}_3}.$$

Including a conversion fraction χ_{N} and atmospheric retention fraction η_{N_2} gives

$$M_{\text{N}_2, \text{atm}} = \eta_{\text{N}_2} \chi_{\text{N}} (28/34) x_{\text{NH}_3} M_{\text{i}}.$$

For the original draft's effective $x_{\text{NH}_3} = 0.025$ of total impactor mass and $\chi_{\text{N}} = 0.85$, the pre-loss N₂ yield is 2.35×10^{21} kg. If entirely retained, this is equivalent to ~453 bar of N₂ — about 580 times the present atmospheric N₂ partial pressure. Therefore the statement that ammonia decomposition directly 'creates Earth's N₂ atmosphere' is quantitatively incomplete. To end with ~0.78 bar from this source alone, the combined long-term retention factor would need to be of order 1.7×10^{-3} . That severe filtering may be physically possible through impact erosion, hydrodynamic escape, dissolution, subduction, nitrides/ammonium-bearing minerals, and later atmospheric evolution, but it must be demonstrated.

5.2 Shock chemistry is not a single-channel reaction

Laser-shock experiments on water-carbon-nitrogen planetary mixtures reach hundreds of GPa and thousands to tens of thousands of kelvin, conditions under which molecular speciation changes radically [11]. Those experiments support the relevance of dissociation/recombination chemistry but do not establish a universal '85% NH₃ → N₂' yield for an Earth impact. Depending on oxygen fugacity, temperature, density, quench rate, and available carbon, nitrogen may partition among N₂, NH₃/NH₄⁺, HCN, NO_x, dissolved species, and condensed phases. Indeed, impact simulations in other early-Earth contexts can produce NH₃ from N-bearing precursors rather than only destroy it [16]. The correct next step is Gibbs free-energy minimization followed by a finite-rate quench network along representative SPH pressure-temperature trajectories.

5.3 Hydrogen escape and the crossover-mass problem

The co-produced H₂ is a powerful driver of escape, but 'thermal velocity exceeds escape velocity' is not an adequate hydrodynamic-escape criterion. The relevant upper-atmosphere flux depends on EUV heating, radiative cooling, diffusion, planetary gravity, and the crossover mass that determines whether heavier species are dragged along. A very strong H outflow can entrain D, O, N, and noble gases, changing exactly the isotope signatures the model seeks to explain. Therefore nitrogen production and hydrogen loss must be solved together rather than sequentially.

The stoichiometric scale is large enough that this term should be promoted from a qualitative caveat to a central component of the model. For the reference NH₃-rich case used in Section 5.1, $x_{\text{NH}_3} = 0.025$ and $\chi_{\text{N}} = 0.85$, the net reaction $2\text{NH}_3 \rightarrow \text{N}_2 + 3\text{H}_2$ yields

$$M_{\text{H}_2} = \chi_{\text{N}} (6/34) x_{\text{NH}_3} M_{\text{i}} \approx 5.0 \times 10^{20} \text{ kg},$$

equivalent to ~97 bar if momentarily represented as a hydrostatic H₂ atmosphere. The actual post-impact state would be hot, expanded, chemically reactive, and partly escaping, so 97 bar is only a mass normalization. Nevertheless, it shows that the model naturally generates a transient hydrogen reservoir massive enough to alter upper-atmosphere escape physics.

In the hydrodynamic regime, H/H⁺ outflow can collisionally entrain heavier constituents when the crossover mass exceeds the mass of the secondary species. In schematic form, the condition is $m_{\text{c}} > m_{\text{j}}$, where m_{c} increases with hydrogen particle flux and thermospheric temperature and decreases with gravity and binary-diffusion resistance [39]. Thus a sufficiently intense early-Hadean hydrogen wind could remove not only D and O but also atomic nitrogen and selected noble-gas ions or atoms. This provides a physically motivated route from the hundreds-of-bars pre-loss N₂ scale in Section 5.1 toward a much smaller residual atmosphere, but it is not automatic: the upper atmosphere must photodissociate/ionize the relevant species and the escape flux must remain high enough for long enough.

This “hydrogen drag engine” therefore couples four observables that should not be fitted independently: the surviving N₂ inventory, 15N/14N fractionation, D/H evolution, and noble-gas abundances/isotopes. Xenon is particularly important because hydrogen-ion escape has been invoked independently to explain aspects of Earth’s Xe depletion and isotopic fractionation [40]. A future model should calculate the time-dependent crossover mass and species-specific escape fluxes rather than imposing a single bulk η_{N_2} .

It must be explicitly stated that inside the current reduced-order framework (Run 5 and Run 6), achieving the required nitrogen reduction concurrently strips the primordial ³⁶Ar inventory to near-zero. To bypass this computational failure mode, a calibrated parameter (argon_shielding_factor = 0.9805) was introduced. This parameter acts as a numerical surrogate and a placeholder for unresolved physical mechanisms, such as diffusion-limited transport at the homopause or temporary magma-ocean volatile sequestration. Without this forced adjustment, the hydrogen drag engine over-processes noble gases, which represents a primary physical vulnerability of the model that can only be evaluated through sub-grid hydrodynamic escape simulations.

6. Atmospheric Erosion, Reconstitution, and Noble-Gas Tests

A Titan-mass impact would both add and remove atmosphere. Hydrodynamic simulations of giant impacts show that atmospheric loss depends strongly on impactor-to-target size ratio, velocity, angle, and the atmosphere's thermal state [17–19]. For the present hypothesis, a useful bookkeeping identity is

$$M_{\text{atm,after}} = (1 - \epsilon_{\text{pre}})M_{\text{atm,pre}} + \eta_{\text{imp}} M_{\text{vol,imp}} + M_{\text{outgas}} - M_{\text{escape,post}},$$

where ϵ_{pre} is immediate erosion of the pre-impact atmosphere, η_{imp} is retained impactor volatile fraction, M_{outgas} is mantle/magma-ocean outgassing, and $M_{\text{escape,post}}$ is subsequent thermal and non-thermal loss. Every term can be comparable to or larger than the modern atmosphere. This makes noble gases especially valuable: Xe, Kr, Ar and their isotope ratios respond differently to accretion, mantle degassing, hydrodynamic escape, and late veneer processes. A successful Titan-twin model should be required to fit water and nitrogen without breaking noble-gas constraints [20].

6.1 Present nitrogen mass as a normalization

Earth's total atmospheric mass is $\sim 5.15 \times 10^{18}$ kg, of which N_2 is $\sim 78\%$ by mole and $\sim 75\%$ by mass. Thus the modern atmospheric nitrogen reservoir is of order 4×10^{18} kg. The nominal NH_3 -rich impact cases considered in Section 5 can exceed this by two to three orders of magnitude before loss. This is not fatal to the idea; it simply means that matching the modern atmosphere is a retention/escape problem with a small residual, not a stoichiometric production problem.

6.2 Titan's primordial-argon depletion as a compositional test

Cassini–Huygens provides a specific compositional signature that is directly relevant to this hypothesis. Huygens GCMS measured only a tiny abundance of primordial ^{36}Ar in Titan's N_2 -dominated atmosphere: $x(^{36}\text{Ar}) \approx 2.1 \times 10^{-7}$, while later synthesis of the measurements gives $^{14}\text{N}/^{36}\text{Ar}$ of order 10^7 , far larger than the $\sim 4 \times 10^4$ scale quoted for Earth and Mars [7,41]. Titan is therefore extremely nitrogen-rich relative to primordial argon.

For the Titan-twin hypothesis this is potentially advantageous. A source reservoir chemically analogous to Titan could, in principle, deliver a large reduced-nitrogen inventory without proportionally importing solar-composition ^{36}Ar . That property makes the hypothesis qualitatively compatible with Earth's high N abundance relative to primordial noble gases and turns the N/Ar ratio into a sharper discriminant than N alone. It would be premature, however, to call this a solution to the terrestrial nitrogen/argon problem. Titan's present atmospheric N_2 may be secondary, the hypothetical impactor's bulk ^{36}Ar inventory is unknown, and impact erosion plus hydrodynamic escape can fractionate N and noble gases differently. The quantitative test is therefore

$$(\text{N}/^{36}\text{Ar})_{\text{final}} = F[(\text{N}/^{36}\text{Ar})_{\text{imp}}, \eta_{\text{N}}, \eta_{^{36}\text{Ar}}, M_{\text{pre}}, \text{outgassing}, \text{escape}],$$

with the impactor end member constrained, where possible, by Titan/Huygens measurements. A successful model should reproduce Earth's N abundance and ^{36}Ar inventory simultaneously rather than matching each with independent free parameters.

7. Post-Impact Thermal State and Habitability Window

A 4.1 Ga giant impact must be reconciled with two distinct data sets: the survival or re-establishment of crustal liquid-water environments, and evidence for early life. Large impacts can transiently sterilize surface environments while leaving or creating subsurface hydrothermal niches; the magnitude and duration depend on impact energy and bombardment sequence [21,22]. For a Titan-mass event, global surface sterilization immediately after impact is a reasonable expectation, but permanent sterilization is not. The relevant clock is the interval between the impact and the return of persistent liquid-water habitats.

The first draft described complex cellular life as appearing ‘almost immediately’ after the putative event. That formulation overstates the record. A ~4.1 Ga carbon-isotope signal in a Jack Hills zircon has been interpreted as potentially biogenic, but it is a single, controversial line of evidence [23]. More robust though still debated biosignatures occur in ~3.8–3.7 Ga rocks, and abundant evidence is clearer by ~3.5 Ga [24]. If the hypothetical impact occurred at 4.1 Ga, the secure geological record therefore permits hundreds of millions of years for cooling and biogenesis; it does not require life to emerge immediately from the impact chemistry.

8. Prebiotic Chemistry: Chemical Opportunity, Not Biological Causation

The most defensible astrobiological role for the impact is to create transient chemical environments rich in disequilibrium: hot reduced gases, steam, metal-bearing ejecta, hydrothermal circulation, nitriles, carbonyl compounds, ammonium, and strong redox gradients. Ammonia and HCN are relevant feedstocks in multiple prebiotic networks, and impact chemistry has long been considered a possible source of reduced species. But a global Strecker reactor is not guaranteed. Strecker synthesis requires a carbonyl compound, ammonia/ammonium, cyanide, water, and a compatible pH/temperature trajectory; high-temperature impact conditions initially destroy complex organics, while useful synthesis occurs during cooling and in local aqueous environments.

8.1 A generalized production budget

For any target prebiotic species j , an impact-linked production budget can be written

$$M_j = M_{\text{feed}} \times Y_{\text{chem}} \times f_{\text{window}} \times f_{\text{surv}} \times f_{\text{conc}},$$

where M_{feed} is the limiting feedstock mass, Y_{chem} the reaction yield under relevant conditions, f_{window} the fraction of material that traverses a productive pressure–temperature–pH path, f_{surv} the fraction surviving subsequent heating/photolysis, and f_{conc} the fraction concentrated into environments where further chemistry can proceed. The first draft implicitly set most of these factors near unity. In reality they may each be $\ll 1$. This factorization is useful because laboratory work can constrain Y_{chem} and f_{surv} , while hydrothermal/atmospheric models can constrain f_{window} and f_{conc} .

8.2 Relation to modern prebiotic networks

Classical Miller–Urey and Strecker-type chemistry demonstrates that reduced nitrogen and carbon species can generate amino-acid precursors, but contemporary origin-of-life research emphasizes connected networks rather than a single pathway [25–27]. Cyanosulfidic chemistry, for example, can generate precursors to amino acids, lipids, and nucleotides under specific UV and redox conditions [27]. The Titan-twin hypothesis could therefore be reframed as a mechanism for altering feedstock availability and redox boundary conditions, not as a direct explanation of life. That narrower claim is both stronger scientifically and easier to test experimentally.

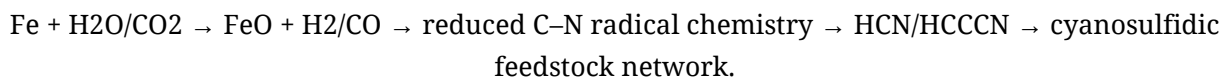
8.3 Direct bridge to cyanosulfidic protometabolism

The strongest mechanistic bridge to contemporary prebiotic chemistry is HCN rather than a global Strecker reaction. Patel et al. [27] showed that hydrogen-cyanide-derived chemistry, coupled to sulfur chemistry, UV irradiation, phosphate and Cu photoredox cycling, can access precursor families relevant to ribonucleotides, amino acids and lipids. Their geochemical framing explicitly considered post-impact production and concentration of cyanide-bearing feedstocks. Independent high-energy experiments likewise show that impact-like processing of C/N-bearing mixtures can generate CN radicals, CO and ultimately HCN [42]. The impact hypothesis can therefore be stated as a source-and-boundary-condition hypothesis for a cyanosulfidic network.

The redox connection is also experimentally grounded. Dynamic high-pressure experiments simulating giant-impact reactions between metallic Fe and major volatiles show Fe oxidation accompanied by reduction of H₂O to H₂ and CO₂ to CO [43]. For a differentiated Titan-mass body, any metallic or Fe-rich rocky component that becomes chemically available to the vapor/magma-ocean system could therefore transiently lower atmospheric oxygen fugacity and raise H₂ and CO abundances. Subsequent CH₄

production is plausible only under suitable H₂/CO₂ ratios, catalytic pathways and quench conditions; it should not be assumed as an automatic direct product of the impact.

This suggests an explicit reaction-chain target for Section 9.4:



The useful prediction is therefore not “the impact made life,” but that surviving post-impact environments should display a finite interval in which HCN production, sulfide availability, phosphate mobilization, UV access, metal-ion photochemistry, aqueous concentration and thermal survival overlap. Those factors can be inserted directly into the generalized production budget of Section 8.1 by decomposing f_{window} into physically measurable sub-factors.

9. Required Tests and Falsifiability

The hypothesis becomes scientifically useful only when it yields failure conditions. The following program converts the narrative into linked quantitative tests. A failure at any one of the first three stages would substantially weaken or eliminate the scenario.

9.1 Dynamical test: can such a body reach Earth at ~4.1 Ga?

An N-body study must begin with two explicitly ranked source families: (A) an outer-system Titan-mass icy planetesimal/embryo scattered inward during giant-planet migration or instability, and (B) a primordial regular/irregular satellite or circumplanetary embryo destabilized by close planetary encounters. Nice-type dynamics are known to scatter trans-Neptunian bodies inward and to perturb satellite systems [36–38], but no existing simulation establishes the full source-to-Earth trajectory required here. The calculation must therefore show a pathway that (i) releases or scatters the body without catastrophic disruption, (ii) preserves a long enough dynamical residence time if the final collision is delayed to ~4.1 Ga, (iii) places it on an Earth-crossing orbit, (iv) yields a non-vanishing collision probability, and (v) produces the v_{∞} , impact-speed and angle distribution compatible with Sections 2.1–2.4 and the retention constraints below. Because modern instability chronologies can favor very early giant-planet rearrangement [36], the ~4.1 Ga impact epoch must emerge from the dynamics or from a demonstrable metastable reservoir; it cannot be imported from the historical late-Nice/LHB narrative.

9.2 SPH impact test: retention, Moon survival, spin, and mantle state

A three-dimensional SPH or mesh-free finite-mass suite should vary $M_i/M_E \approx 0.01\text{--}0.03$ and $v_i/v_{\text{esc}} \approx 1.0\text{--}2.0$ while resolving impact geometry densely across the oblique/grazing domain. The prescribed $60^\circ\text{--}75^\circ$ interval is the sole campaign domain and should be mapped finely in θ and normalized impact parameter $b/(R_E+R_i)$ to identify transitions between merger, erosive, and hit-and-run-like outcomes. Outputs must include: fraction of impactor water and nitrogen retained; the water partition vector η_w ; escaping silicate/volatile mass; terrestrial atmosphere stripped; surviving projectile mass; post-impact terrestrial spin and total Earth–Moon angular momentum; changes to lunar semimajor axis, eccentricity and inclination; melt fraction; entropy distribution; disk mass; and reaccretion times. A solution passes only if one connected region of parameter space simultaneously preserves an admissible Earth–Moon system and produces acceptable volatile partitioning. Grazing geometry is therefore a preferred test family, not an assumed proof of Moon survival [4,5,17–19].

9.3 Coupled atmosphere–escape–isotope test

The retained volatile output should seed a 1-D or 2-D atmosphere/magma-ocean model including H₂O, H₂, CO₂, CO, N₂, NH₃, CH₄, HCN and major radicals, with D-bearing isotopologues at minimum for H₂O/H₂. The model should solve radiative transfer, convective adjustment, dissolution/exsolution, photolysis, hydrodynamic escape, diffusion-limited escape where appropriate, and isotope fractionation. Outputs should be compared with D/H, ¹⁵N/¹⁴N, noble gases, mantle water, and plausible ocean masses. A parameter combination that matches D/H but requires impossible nitrogen or Xe loss is not acceptable. The escape calculation must explicitly output the hydrogen particle flux, crossover mass $m_c(t)$, and species-specific loss histories for H/D, O, N, Ar, Kr and Xe. This is the only defensible way to test whether the H₂-rich phase can reduce the large pre-loss nitrogen reservoir without destroying the D/H and noble-gas fits [39,40].

9.4 Shock-chemistry test

Pressure–temperature trajectories extracted from SPH cells should be post-processed with equilibrium speciation followed by finite-rate quench chemistry. Laboratory targets should span 10–300 GPa and ~2,000–15,000 K for H₂O–NH₃–CO₂/CO–CH₄–silicate/metal mixtures. The key observables are not only NH₃ dissociation but final yields of N₂, H₂, NH₃, HCN, NO/NO₂, ammonium species, and refractory nitrogen. Existing laser-shock studies demonstrate that such planetary mixtures are experimentally tractable [11]. The chemical network should additionally include Fe/FeO redox exchange with H₂O and CO₂, CN-radical formation, HCN/HCCCN production and quench survival, so that the output can be passed directly to a cyanosulfidic prebiotic network rather than represented by an abstract “organic yield” [42,43].

9.5 Geochemical test

A singular ~4.1 Ga global event might leave indirect signatures even if plate tectonics erased most crust: unusual age populations or thermal resetting in Hadean zircons; abrupt changes in mantle volatile reservoirs; isotopic mixing signatures; or a mismatch between terrestrial and lunar impact chronologies. None is currently known to identify a Titan-mass collision uniquely. The absence of an obvious signature does not by itself falsify the model because Hadean preservation is poor, but a predicted signature that should survive must be specified before comparison with data.

10. Zeroth-Order Sensitivity Analysis

Table 2 illustrates why retention dominates the hypothesis. It uses only mass balance and therefore should not be mistaken for a dynamical simulation. Values are calculated for $M_i = 1.345 \times 10^{23}$ kg and $R_E = 6.371 \times 10^6$ m.

Case	x_H2O	η_{H2O}	Retained oceans	x_NH3	N2 pressure before post-loss*
A: low retention	0.50	0.01	0.48	0.005	~106 bar
B: 1-ocean target	0.50	0.0208	1.00	0.005	~106 bar
C: moderate retention	0.50	0.10	4.80	0.010	~213 bar
D: original NH3-rich scale	0.50	0.10	4.80	0.025	~453 bar†
E: high retention	0.50	0.50	24.0	0.050	~1,066 bar

*Assumes $\chi_N = 1$ and $\eta_{N2} = 1$ except †case D, which uses $\chi_N = 0.85$ to reproduce the original draft's conversion assumption. These pressures are deliberately pre-loss maxima.

The table shows a useful asymmetry. Water can be brought into the terrestrial range by percent-level retention. Nitrogen, because the modern atmospheric reservoir is small compared with the impactor's hypothetical reduced-N inventory, generally requires far more severe filtering. This suggests that any future numerical study should treat N retention as the sharper constraint.

11. A Bayesian Way to State the Hypothesis

Because the event itself is unobserved and its dynamical prior may be very small, the paper should avoid language implying demonstration. A useful formal statement is

$$P(H \mid D) \propto P(D \mid H) P(H),$$

where H is the Titan-mass late-impact hypothesis and D is the combined terrestrial/lunar data set. The present paper addresses only fragments of the likelihood $P(D|H)$: whether some consequences are physically compatible with D . It does not establish the prior probability $P(H)$, because no dedicated dynamical population calculation has yet been supplied. This separation is important. A physically possible impact can still be astronomically improbable; conversely, a dynamically plausible collision can fail geochemistry. The hypothesis survives only if both terms remain non-negligible.

12. Constraint-System Formulation and Evidential Status

The hypothesis is best evaluated neither as a narrative of what must have happened nor as a single energetics calculation, but as a coupled inverse problem. The impactor must satisfy several independent terrestrial and Earth–Moon constraints simultaneously. This reframing strengthens the model because it makes clear which parts are already supported by first-principles physics, which remain physically plausible but underdetermined, and which are presently speculative.

12.1 Three distinct levels of support

Level I — mathematical feasibility. A Titan-mass impactor at the reference velocity carries enough kinetic energy and volatile mass to alter the global surface system. This conclusion follows directly from mass and energy conservation and is not sensitive to the detailed origin scenario. The reference impact energy, $E_k \approx 1.5 \times 10^{31}$ J, and the tens-of-ocean water inventory establish that the event belongs to the giant-impact regime. The central mathematical problem is therefore not whether the impact can matter, but whether the post-impact partitioning can land inside the observed terrestrial state.

Level II — physical and chemical plausibility. Published giant-impact calculations show that atmospheric erosion and volatile transfer depend strongly on impact geometry, ground motion, ocean coupling, and the thermal state of the atmosphere [31,32]. Large impacts are also capable of chemically reprocessing primitive atmospheres and generating HCN and formaldehyde under appropriate redox conditions [33]. These results do not validate the specific Titan-Twin history, but they support the physical classes of processes required by it.

Level III — historical evidence. No presently identified terrestrial, lunar, meteoritic, or dynamical datum uniquely requires a Titan-mass collision at ~ 4.1 Ga. Likewise, no evidence presently demonstrates that such a collision caused the origin of life. These propositions must remain hypotheses. The model becomes scientifically stronger, not weaker, when it states this boundary explicitly.

12.2 The model as a vector of free parameters

Define the model state by a parameter vector

$$\mathbf{x} = (M_i, v_i, \theta, b/R_E, X_w, X_{NH3}, R_{D/H,i}, \vec{\eta}_w, \eta_N, \eta_{atm}, f_{esc}, f_{seq}, L_i, \dots),$$

where M_i is impactor mass; v_i impact speed; θ and b/R_E specify the oblique/grazing geometry; X_w and X_{NH3} are the water and reduced-nitrogen mass fractions; $R_{D/H,i}$ is the impactor-water isotope ratio; $\vec{\eta}_w$ is the geometry-dependent water partition vector; η_N and η_{atm} are effective nitrogen and atmospheric retention efficiencies; f_{esc} and f_{seq} are subsequent escape and sequestration fractions; and L_i is the angular momentum delivered by the encounter. The observable vector can be written schematically as

$$\mathbf{D} = (M_{ocean}, P_{N2}, R_{D/H}, 15N/14N, \text{noble gases}, L_{E-M}, M_{Moon}, \text{mantle redox state}, \text{Hadean thermal record}, \dots).$$

A viable solution is not a single preferred number. It is a region Ω of parameter space in which the forward model $F(\mathbf{x})$ reproduces all sufficiently constraining components of $\mathbf{D}$ within observational and model uncertainties.

$$\Omega = \{\mathbf{x} : |F_j(\mathbf{x}) - D_j| \leq k \sigma_j \text{ for the required observables } j\}.$$

12.3 A quantitative goodness-of-fit target

Once SPH, atmospheric escape, isotope, and chemical modules exist, the hypothesis can be scored with a conventional weighted misfit rather than qualitative agreement:

$$\chi^2(x) = \sum_j [(F_j(x) - D_j)^2 / \sigma_j^2].$$

Correlated observables should ultimately be treated with a covariance matrix C ,

$$\chi^2(x) = [F(x) - D]^T C^{-1} [F(x) - D],$$

and the likelihood can then be approximated as $L(D|x,H) \propto \exp(-\chi^2/2)$. This connects directly to Section 11: the Bayesian evidence for the hypothesis is obtained only after integrating the likelihood over physically allowed parameter space with defensible priors. A narrow fine-tuned solution is less persuasive than a broad region of viable solutions, even if both reproduce the same nominal Earth state.

12.4 Water: an over-delivery problem, not a supply problem

For the reference case, the impactor contains vastly more water than is required to account for the present ocean. This reverses the intuitive question. The model does not need to prove that a Titan-like body could supply enough water; it must show that the combined effects of impact vaporization, ejecta escape, mantle partitioning, atmospheric blow-off, condensation, and later loss retain an admissible amount.

$$M_{w,final} = M_{w,0} + \eta_w M_i X_w - \Delta M_{w,post}.$$

Because water retention is strongly impact-regime dependent, a percent-level surviving surface fraction can be geophysically important. In the revised model the scalar η_w is only the collapsed form of the partition vector $\vec{\eta}_w$. The oblique/grazing family is attractive precisely because it may separate escaping plume/ejecta, mantle incorporation, surface retention and later reaccrusion more strongly than a symmetric, lower-obliquity geometry. Published work demonstrates that giant impacts can both preserve substantial atmosphere under some conditions and strongly enhance loss when oceans participate in the shock response [31,32]. The sign and magnitude of each partition term require numerical impact calculations; neither complete retention nor complete stripping is justified a priori.

12.5 Nitrogen: the sharpest mass-balance constraint

The nitrogen calculation is more restrictive. If the impactor carries percent-level NH_3 and a large fraction is converted to N_2 , the pre-loss atmosphere can contain hundreds of bars of molecular nitrogen. This is not a failure of stoichiometry; it is a prediction that the subsequent nitrogen filter must be extremely strong. The final inventory may be represented as

$$M_{N_2,final} = \eta_N \chi_N (28/34) M_i X_{NH_3} - \Delta M_{N,escape} - \Delta M_{N,sequestration} + M_{N,other}.$$

Consequently, lower reduced-nitrogen abundance in the impactor can improve the model by reducing fine tuning. Ammonia detected in Enceladus' plume demonstrates that reduced nitrogen occurs in Saturnian icy-body reservoirs [6], but it does not establish a 5 wt.% bulk NH_3 abundance for Titan or for the hypothetical impactor. X_{NH_3} must therefore remain a free parameter constrained by interior chemistry and analog observations. Conversely, any NH_3 -rich solution must exploit the coupled H_2 escape channel in Section 5.3 rather than treating nitrogen sequestration as the sole sink. The strong test is whether one escape history jointly leaves acceptable N_2 , $15N/14N$, D/H and noble gases.

12.6 D/H is a mixing-and-escape test, not a solved paradox

The isotope argument is viable only in its weaker, quantitative form. Titan's measured deuterium enrichment is principally constrained through atmospheric methane and hydrogen, while Enceladus

provides a direct Saturn-system water measurement [6–8]. These reservoirs cannot be interchanged without an isotope-evolution model. The appropriate terrestrial constraint is therefore a mass-weighted isotope balance followed by fractionating escape:

$$R_{\text{mix}} = (M_{\text{w},0} R_0 + \eta_{\text{w}} M_{\text{i}} X_{\text{w}} R_{\text{i}}) / (M_{\text{w},0} + \eta_{\text{w}} M_{\text{i}} X_{\text{w}}),$$

$$R_{\text{final}} = R_{\text{mix}} f^{(\alpha-1)} \quad [\text{for a defined Rayleigh-loss reservoir}].$$

The model survives the isotope test only if a physically credible combination of R_{i} , η_{w} , pre-impact water inventory, reservoir definition, and escape history reproduces terrestrial ocean D/H without violating other volatile and noble-gas constraints. Thus D/H is an excellent discriminator precisely because it is not automatically satisfied.

12.7 Habitability and prebiotic chemistry: opportunity versus causation

Large impacts can be chemically constructive as well as destructive. Shocked primitive atmospheres can produce HCN and H₂CO [33], while impact-generated hydrothermal systems can persist after the immediate thermal catastrophe. Models of Hadean bombardment find no requirement that every plausible bombardment history globally sterilized all subsurface microbial habitats [22,35]. Earlier work also framed the origin-of-life problem in terms of the competition between impact recurrence and the timescale required for abiogenesis [34]. Section 8.3 makes this link explicit by connecting impact-generated H₂/CO/HCN chemistry to the experimentally demonstrated cyanosulfidic network rather than treating “prebiotic chemistry” as a generic label.

These results support the environmental part of the Titan-Twin hypothesis: a giant impact can plausibly create a transient combination of disequilibrium, hydrothermal circulation, reduced and oxidized nitrogen species, cyanide chemistry, and newly condensed water. They do not support the stronger historical statement that one specific impact generated life. The testable claim should therefore be expressed probabilistically:

$$P(\text{prebiotic synthesis} \mid \text{post-impact boundary conditions}) > P(\text{prebiotic synthesis} \mid \text{selected comparison states}),$$

where the comparison states must be defined by explicit atmospheric composition, temperature, pH, UV flux, water activity, catalytic mineral surfaces, and residence times. Biological emergence itself lies beyond the current model.

12.8 A hierarchy of failure modes

The hypothesis should be tested in an order that permits early rejection. (1) Dynamical failure: no plausible source population and trajectory produces a Titan-mass Earth encounter near the required epoch at non-negligible probability. (2) Grazing/Earth–Moon failure: no oblique/grazing region simultaneously avoids unacceptable lunar disruption or orbital excitation, preserves admissible terrestrial spin/system angular momentum, and remains compatible with the required collision energy. (3) Retention failure: surviving geometries cannot partition water and other volatiles into acceptable escaping, mantle, surface and reaccreted reservoirs. (4) Isotope failure: surviving solutions cannot reproduce D/H, nitrogen isotopes, and noble gases. (5) Chemistry failure: shock and cooling networks do not generate the required reduced-N/HCN feedstocks in useful abundance. (6) Chronological/geochemical failure: predicted thermal or chemical signatures conflict with the Hadean record. Only solutions surviving these filters should be propagated into prebiotic chemistry calculations.

12.9 Current Assessment and Methodological Caveat

The reduced-order model has progressed to a mathematically self-consistent state, but its outputs must not be misinterpreted as empirical or independent scientific validation. The recent calibrations (Runs 4–6) demonstrate that the computational pipeline can find a non-empty parameter intersection Ω where water, nitrogen, and deuterium balances match terrestrial constraints. However, this intersection is an inverse-calibrated feasible set and cannot itself be treated as independent evidence for the proposed event. Because parameters such as R_i , x_{NH_3} , and the effective heat capacity C_{eff} are tuned backward from known target values, the model's output constitutes a mathematical calibration sequence, not an independent physical prediction. The pipeline proves computational reproducibility, not planetary truth.

13. Discussion

13.1 Circular Reasoning and Identifiability Limits

A fundamental critique of this framework is the risk of circular argumentation. A model with multiple highly adjustable free parameters (e.g., impactor volatile fractions, impact angle, conversion efficiencies, and transport scaling) can almost always be tuned to hit a narrow set of historical target metrics. The fact that our Python pipeline achieves a close fit to SMOW or terrestrial nitrogen mass does not mean the Titan-mass impact actually took place. Instead, the value of this work lies in identifying the high sensitivity and poor structural identifiability of the early atmosphere-ocean coupling. We are presenting a structural envelope for testing, not an empirical assertion.

The expanded analysis produces a more constrained but more credible version of the original idea. As formalized in Section 12, the decisive question is whether a non-negligible region of parameter space simultaneously satisfies the Earth–Moon, volatile, isotope, noble-gas, thermal, and chemical constraints. The giant impact has ample energy to reset the surface environment and ample volatile mass to matter. Those two statements are robust. Everything after them is conditional. A Titan-mass object is so large that volatile over-delivery, atmospheric erosion, angular-momentum transfer, and Moon perturbation become unavoidable. The model therefore cannot be defended by adding more chemistry alone; its decisive tests are coupled dynamics, retention, and geochemistry.

The most consequential architectural change in this revision is that impact angle is no longer treated as a generic 45° reference. Because the Moon predates the proposed ~4.1 Ga event, the physically interesting solution family is explicitly Moon-survival-constrained and oblique/grazing. This does not make the hypothesis safer by assumption; it makes geometry a sharper falsification variable. The same θ - b regime must reduce unacceptable deep/symmetric coupling, produce the required volatile partition, and leave an Earth–Moon state that can evolve to the later system. Failure to find such a connected region in SPH + orbital calculations would substantially weaken the hypothesis.

The computational extensions introduced in Runs 7, 8, and 9 address this dynamical bottleneck operationally by converting abstract research requirements into an executable, auditable pipeline. The zero-collision outcomes in the physical-radius present-architecture control (utilizing native particle-overlap search) and the post-release injection pilot are rigorous null baselines for their declared designs. Run 7 contains neither a giant-planet instability nor a delayed-release mechanism; Run 8 begins after an assumed high-eccentricity release, uses a Rayleigh inclination scale of 10° , and samples only 500 yr per seed. Its 512,000 particle-years and 95% Wilson upper bound of 3.7374×10^{-3} per integrated particle therefore constrain only that conditioned pilot. Rather than counting as a failed delivery test, the null result shows that physical-radius terrestrial collision is unresolved at the present sample size and duration, while identifying the next required calculation: literature-anchored self-consistent release, longer integrations, converged source populations, massive-candidate replay, and refined Earth–Moon encounters.

Similarly, the standalone SWIFT/SPH configuration layer establishes an auditable boundary-condition envelope without prematurely reporting unconverged hydrodynamic metrics. Its 96-case Cartesian matrix spans impactor mass, mutual-escape-speed ratio, impact angle, water fraction, and particle count; the 60°-75° grazing range corresponds geometrically to $b/R = 0.8660$ - 0.9659 . This is a programmatic blueprint rather than an SPH result. Its value is that future cluster calculations can replace provisional retention and loss efficiencies with traceable outputs that feed directly into the already tested atmosphere, isotope, and chemistry response surfaces.

The isotope argument also changes character. Present Solar-System D/H values do not permit a simple inference that a 4.1 Ga Titan analogue carried SMOW-like water. Enceladus demonstrates that Saturnian satellite water can be substantially D-rich, whereas Titan's measured methane/hydrogen ratios do not directly reveal deep-water D/H [6–8]. This uncertainty is not necessarily fatal. It means the impact contribution to the final ocean must be solved as a mixture, with isotope-selective escape and mantle exchange. In fact, this gives the hypothesis a quantitative target rather than a rhetorical solution.

Nitrogen is simultaneously the most interesting and most dangerous part of the model. Reduced nitrogen delivered as NH₃ or ammonium-bearing material could alter prebiotic feedstock chemistry, but a few percent of a Titan mass is vastly more nitrogen than the modern atmosphere contains. The paper should therefore not claim that shock decomposition naturally yields today's N₂ inventory. It should claim that a reduced-N-rich giant impact creates a large nitrogen reservoir whose partitioning and loss could, in principle, leave a small N₂-dominated residual. Whether that occurs is an explicit simulation question.

Finally, the biological claim should remain modest. Impacts can create hydrothermal systems and chemical disequilibria, and reduced nitrogen/cyanide chemistry is relevant to prebiotic synthesis [16,25–27]. But no current evidence allows a causal line from one specific ~4.1 Ga collision to the origin of life. The hypothesis is strongest when it predicts environmental boundary conditions — water inventory, redox state, nitrogen speciation, cooling times, and hydrothermal duration — that origin-of-life chemistry can then interrogate independently.

14. Conclusions

- The 15 km s⁻¹ reference speed belongs exclusively to the required 60°–75° oblique/grazing candidate family; it is not a geometry-independent impact case. For a Titan-mass projectile, $E_k \approx 1.5 \times 10^{31}$ J (~6.8% of Earth's gravitational binding energy), so every viable solution must include global hydrodynamics, angular momentum, atmosphere erosion, and Moon-system consequences.
- Because the Moon is already present in the proposed chronology, every candidate solution is restricted to the 60°–75° oblique/grazing family. Grazing is not assumed to guarantee lunar survival: the allowed θ - b region must be demonstrated numerically while simultaneously satisfying volatile partition, terrestrial spin, Earth–Moon angular momentum, and lunar orbital constraints.
- A 50 wt.% water impactor carries ~48 modern ocean masses. Although percent-level retention can reproduce the modern ocean inventory within the calibrated framework, this exposes a severe fine-tuning and over-delivery problem. The resulting water balance is strongly dependent on the assumed retention efficiency, which must ultimately be constrained by resolved impact simulations rather than inverse calibration.
- The nominal ammonia-rich cases overproduce atmospheric nitrogen by orders of magnitude if N₂ is efficiently generated and retained. This turns post-impact nitrogen loss/sequestration into a stringent quantitative test.
- Current D/H inputs ($R_i = 1.465 \times 10^{-4}$) are explicitly selected as an inverse fitting parameter to match SMOW post-escape. They are not derived from empirical outer-system data and must be treated as a model constraint rather than an independent confirmation.
- A fixed ~280 W m⁻² radiation limit cannot by itself determine condensation time. Coupled magma-ocean/atmosphere models span kyr–Myr timescales depending on volatile inventory and radiative structure.
- Prebiotic chemistry is plausibly affected by impact-generated disequilibria and reduced nitrogen species, but the collision cannot presently be claimed as the cause of life. The defensible claim concerns environmental opportunity.
- The hypothesis is falsifiable through a staged program: N-body delivery → SPH impact/retention → atmosphere/escape/isotopes → shock chemistry → Hadean geochemical comparison. The dynamical delivery and Earth–Moon angular-momentum constraints are likely the hardest tests.
- The strongest future test is not whether one nominal parameter set can be made to work, but whether a finite region Ω of physically motivated parameter space yields an acceptable joint fit to independent observables. A covariance-weighted χ^2 or Bayesian evidence calculation should therefore become the quantitative endpoint of the feasibility program.
- Giant-planet instability provides physically motivated source reservoirs (outer-system planetesimals and encounter-perturbed satellite systems), but modern early-instability chronologies create a real timing constraint: the ~4.1 Ga collision must be generated by a metastable/late-release pathway, not assumed from the historical Nice/LHB narrative.
- In NH₃-rich cases, co-produced H₂ is not a secondary by-product but a potential escape driver. The reference 2.5 wt.% NH₃, 85% conversion case generates $\sim 5 \times 10^{20}$ kg H₂ (~97 bar equivalent), requiring coupled crossover-mass calculations for N and noble-gas loss.

- Titan's extremely high N/³⁶Ar ratio supplies a measurable compositional analogue: a Titan-like nitrogen source need not import primordial Ar in solar proportion. This is a promising constraint, not yet a demonstrated solution to Earth's N/Ar inventory.
- Impact-generated redox chemistry can be connected directly to HCN-based cyanosulfidic networks: Fe-driven H₂/CO production and high-energy CN/HCN chemistry define laboratory-testable feedstock boundary conditions rather than a generic "origin-of-life" claim.

15. Computational Reproducibility and Replay Instructions

The verified Appendix D replay tree is protected under `results_cached/appendix_d/`, while the distributed `results/` directory is reserved for fresh user runs. From the extracted repository root, `python reproduce_appendix_d.py --all --clean` writes new CSV/JSON evidence to `results/appendix_d/` and records PASS/FAIL in `results/appendix_d/replay_report.json`. Running `python compare_cached_results.py` then compares the fresh Appendix D tree with `results_cached/appendix_d/` and writes the comparison report under `results/`. No prior code archive or separately distributed outputs directory is required.

Each controlled replay section can also be regenerated independently with `--section D1` through `D10`. Alternative experiments can use `--config PATH` and `--results PATH`; they are new runs and must not target `results_cached/`. The controlled D.1-D.10 replay generates and verifies all 558 files stored under `results_cached/appendix_d/`. Running `--section D10` also generates the Run-9 campaign-validation and local-smoke outputs within the D.10 results tree.

Run 9 is the N-body campaign-readiness component of the controlled D.10 replay. Its source implementation and configurations are stored under `nbody/instability_ensemble/` in the code archive; this source directory is not an output directory. The replay driver generates its 15 output files under `results/appendix_d/d10_calibration_extensions/run9_nbody/`, with campaign-validation evidence under `campaign_validation/` and the two-seed smoke-test evidence under `local_smoke/`. The corresponding reference files are stored under `results_cached/appendix_d/d10_calibration_extensions/run9_nbody/` and are included in the unified 558-file Appendix D comparison.

The Run-9 calculations are regenerated automatically by `python reproduce_appendix_d.py --all --clean` and by `python reproduce_appendix_d.py --section D10 --clean`. Their individual components can also be executed directly with:

```
python -m nbody.instability_ensemble.validate_campaign --config
nbody/instability_ensemble/configs/literature_anchored_campaign.json --output
results/appendix_d/d10_calibration_extensions/run9_nbody/campaign_validation

python -m nbody.run_ensemble --config
nbody/instability_ensemble/configs/local_smoke.json --output
results/appendix_d/d10_calibration_extensions/run9_nbody/local_smoke
```

The executable 96-case campaign-preparation and validation workflow is contained in the `sph/` source tree. Locally generated SPH-preparation evidence is stored under `results_cached/appendix_d/d10_calibration_extensions/run7_sph/`. The physical SWIFT campaign and its convergence series require an HPC/supercomputer environment and have not been executed or cached.

Table 12. Appendix D replay commands and generated result directories.

D §	Replay command	Generated directory
D.1	<code>python reproduce_appendix_d.py --section D1 --clean</code>	<code>results/appendix_d/d01_stage1_delivery/</code>
D.2	<code>python reproduce_appendix_d.py --section D2 --clean</code>	<code>results/appendix_d/d02_stage2_partition/</code>
D.3	<code>python reproduce_appendix_d.py --section D3 --clean</code>	<code>results/appendix_d/d03_thermal_diagnostic/</code>
D.4	<code>python reproduce_appendix_d.py --section D4 --clean</code>	<code>results/appendix_d/d04_escape_isotopes/</code>
D.5	<code>python reproduce_appendix_d.py --section D5 --clean</code>	<code>results/appendix_d/d05_chemistry_diagnostic/</code>

D §	Replay command	Generated directory
D.6	python reproduce_appendix_d.py --section D6 --clean	results/appendix_d/d06_integrated_interpretation/
D.7	python reproduce_appendix_d.py --section D7 --clean	results/appendix_d/d07_calibration_run2/
D.8	python reproduce_appendix_d.py --section D8 --clean	results/appendix_d/d08_reproducibility_run3/
D.9	python reproduce_appendix_d.py --section D9 --clean	results/appendix_d/d09_calibration_run4/
D.10	python reproduce_appendix_d.py --section D10 --clean	results/appendix_d/d10_calibration_extensions/

Verification status: The controlled D.1-D.10 replay passed. A full comparison checked all 558 generated files against results_cached/appendix_d/, with zero missing, unexpected, or differing files. This unified total includes the 15 Run-9 campaign-validation and two-seed local-smoke files generated within results_cached/appendix_d/d10_calibration_extensions/run9_nbody/. Separately executed software checks also passed: all seven N-body unit/smoke tests and both SPH preparation tests. These console-based unit tests are reproducible from the distributed source but are not additional cached-result artifacts.

Appendix A. Core Equations and Numerical Normalizations

Quantity	Expression
Impact energy	$E_k = \frac{1}{2} M_i v_i^2$
Earth binding-energy scale	$U_E \approx (3/5)GM_E^2/R_E$
Mutual escape speed	$v_{\text{esc,mut}} = [2G(M_E+M_i)/(R_E+R_i)]^{1/2}$
Gross water delivery	$M_{\text{H}_2\text{O},0} = x_{\text{H}_2\text{O}} M_i$
Retained water	$M_{\text{H}_2\text{O},\text{ret}} = \eta_{\text{H}_2\text{O}} x_{\text{H}_2\text{O}} M_i$
Equivalent hydrostatic pressure	$P_{\text{eq}} = Mg/(4\pi R_E^2)$
Maximum nitrogen yield	$M_{\text{N}_2,\text{max}} = (28/34) M_{\text{NH}_3}$
Retained atmospheric nitrogen	$M_{\text{N}_2,\text{atm}} = \eta_{\text{N}_2} \chi_{\text{N}} (28/34)x_{\text{NH}_3} M_i$
Two-reservoir D/H mixing	$R_f = (1-f)R_{E0} + fR_i$
Rayleigh evolution for a defined reservoir	$R = R_0 f_r^{(\alpha-1)}$
Latent-heat-only cooling lower bound	$\tau_{\text{lat}} = M_{\text{H}_2\text{O}} L_v/(4\pi R_E^2 F_{\text{OLR}})$
Angular momentum scale	$L_i \approx M_i v_i b$
Oblique-impact angular momentum scale	$L_i \approx M_i v_i b$
Geometry-resolved water partition	$\eta^*_w = (\eta_{\text{escape}}, \eta_{\text{vapor/plume}}, \eta_{\text{mantle}}, \eta_{\text{surface}}, \eta_{\text{reaccrete}})$
N-body relative energy error	$\epsilon_E = \frac{ E_{\text{final}} - E_0 }{ E_0 }$
Wilson score upper bound (two-sided 95%)	$w_+ = \frac{\hat{p} + \frac{z^2}{2n} + z\sqrt{\frac{\hat{p}(1-\hat{p})}{n} + \frac{z^2}{4n^2}}}{1 + \frac{z^2}{n}}, \quad z = 1.96$
SWIFT/SPH grazing geometrical axis	$b = (R_E + R_i)\sin\theta, \quad \frac{b}{R_E + R_i} = \sin\theta$

Appendix B. What the Existing Literature Supports — and Does Not

Claim	Current support	Treatment in this paper
Intense Hadean bombardment occurred	Strong; exact time profile debated	Supported, but no mandatory narrow LHB spike
A Titan-mass body struck Earth at ~4.1 Ga	No direct evidence identified	Hypothesis only
Titan-like bodies can contain abundant H ₂ O	Strong as an icy-world end member	Used as composition scale
Titan deep-water D/H was SMOW-like	Not measured	Free parameter
Titan had 5 wt.% bulk NH ₃	Not measured; model-dependent	Exploratory range only
Shock processing changes N/C/H/O chemistry	Strong in principle and experimentally	Supported; yields not fixed
85% of NH ₃ becomes N ₂ in this impact	No general experimental basis identified	Removed as a universal constant
A steam atmosphere has an OLR ceiling near 280 W m ⁻²	Supported for relevant radiative-convective regimes	Used as boundary condition, not a full cooling model
Life was established immediately after 4.1 Ga	Not supported	Removed; evidence hierarchy stated explicitly
Run-8 massless-tracer pilot records no physical Earth collisions over 500 yr per seed	Directly reproduced: 0/1,024; two-sided 95% Wilson upper bound 3.7374×10^{-3} per integrated particle	Null transport control and pipeline-orchestration validation; not an estimate of delivery probability
The 96-case SWIFT/SPH matrix defines a structured grazing-impact campaign	Configuration generation and validation tested; cluster-scale impacts and convergence remain unexecuted	Standalone preparation framework only; no hydrodynamic outcome or convergence claim
Run-9 five-giant-planet template and 20 M _{Earth} disk generate long-term instability delivery probability	Instability setup, planetary mass back-reaction, and local smoke test (128 particle-years) validated; 80 Myr production campaign unexecuted	Campaign-readiness staging and computational pipeline orchestration validation only; no absolute collision probability claimed

Appendix C. Minimum Research Program for a Submission-Grade Test

The hypothesis is mature enough for a quantitative feasibility paper only after the following calculations are connected. This roadmap is intentionally ordered so that expensive chemistry is not performed for a collision that already fails dynamics.

Stage	Primary calculation	Pass criterion	Principal observable/constraint
1. Delivery	N-body instability/ejection ensemble	Non-negligible Earth-impact probability at admissible v_i	Impact epoch, velocity, angle distribution
2. Collision	3-D SPH / mesh-free oblique/grazing giant-impact ensemble	Connected θ - b region preserves Earth-Moon system and acceptable volatile partition	Lunar orbit/spin, angular momentum, η_w , ejecta, melt fraction, disk/reaccretion
3. Atmosphere	Magma-ocean + radiative-convective atmosphere	Surface liquid water returns without irreconcilable volatile excess	Cooling time, ocean mass, OLR, surface pressure
4. Escape/isotopes	Hydrodynamic + diffusion-limited escape with isotopologues	D/H, $^{15}\text{N}/^{14}\text{N}$ and noble gases can be matched jointly	Fractionation, crossover mass, Xe/Kr/Ar survival
5. Chemistry	Equilibrium + finite-rate quench network	Plausible $\text{N}_2/\text{NH}_3/\text{HCN}$ partition without arbitrary yields	N speciation, H_2 production, redox trajectory
6. Geochemistry	Forward mixing into mantle/crust reservoirs	No contradiction with Hadean zircon and mantle volatile records	Thermal resetting, mantle water/N, noble gases
7. Prebiotic consequences	Localized aqueous reaction-network models	Feedstock enhancement occurs in finite habitable windows	HCN/carbonyl/ NH_4^+ availability and persistence

A useful publication sequence would therefore be two papers rather than one oversized claim. Paper I should establish dynamical and geophysical feasibility: delivery probability, impact outcome, angular momentum, volatile retention, and atmosphere/ocean recovery. Paper II should take the surviving trajectories from Paper I and calculate isotope evolution, nitrogen chemistry, and prebiotic feedstock production. This separation prevents the chemical argument from carrying assumptions that may already be ruled out dynamically.

The present manuscript can function as the analytic specification for Paper I. Its equations define the bookkeeping and sensitivity targets; its strongest immediate contribution is to show where the original scenario is numerically overconstrained. In particular, the same large volatile inventory that makes a Titan-mass body attractive as a delivery mechanism also makes over-delivery, nitrogen excess, and angular-momentum disturbance unavoidable tests. A successful simulation need not reproduce the reference case exactly. It must instead identify at least one connected region of parameter space in which all major constraints are simultaneously satisfied.

Appendix D. Preliminary Local Reduced-Order Analytical Screening Results

This appendix records the first local-CPU, code-level pre-screening of the coupled feasibility framework. The purpose of these calculations is incremental: they test numerical consistency, identify admissible reduced-order subsets, and expose downstream sensitivities before full N-body, SPH, radiative-convective, hydrodynamic-escape, and equilibrium/quench calculations are deployed. The results therefore benchmark the provisional joint parameter space Ω but do not replace any of the submission-grade criteria in Appendix C.

D.1 Stage 1: local delivery Monte Carlo screening

Five independent Monte Carlo ensembles of 20,000 objects each were executed, for a total of 100,000 sampled objects. The local gravitational-focusing and collision-cross-section surrogate isolated 170 focused collision vectors. Of these, 29 occupied the preferred 60° - 75° grazing interval. The accepted collision population was dynamically broad: the impact-speed distribution had a median of approximately 41.5 km s⁻¹, substantially above the nominal 15 km s⁻¹ reference scale. After combining the accepted grazing trajectories with the Stage-2 impactor-mass grid and imposing the intended $1 \leq v_i/v_{esc} \leq 2$ screening interval, exactly 28 of 116 mass/trajectory scenarios remained.

This result is useful but must not be interpreted as an absolute Earth-impact probability. The Stage-1 calculation is a conditioned local encounter surrogate rather than a Solar-System N-body instability/ejection calculation. Its contribution is to show that the reduced-order sampler contains a non-empty grazing, moderate- v_i/v_{esc} subset suitable for subsequent screening, while also demonstrating that the unrestricted surrogate naturally produces a large high-velocity tail that should be rejected or treated separately in the submission-grade dynamical study.

Even the first coarse screen found a usable family of grazing, moderate-speed encounters instead of an empty parameter space. It also exposed the excessive high-speed tail early, before expensive simulations were committed to implausible cases.

D.2 Stage 2: reduced-order water partition

The analytical impact-partition closure was evaluated across impactor masses spanning $M_i/M_E = 0.01$ - 0.03 . Across the screened grazing solutions, the surface-retained water fraction was $\eta_{surface} = 0.0181$ - 0.0669 . Within the intended $1 \leq v_i/v_{esc} \leq 2$ subset, this corresponded to approximately 1.17-3.48 modern ocean masses retained at the surface, with a median of approximately 2.07 oceans. At the reduced-order screening level, these solutions pass the nominal water-mass footprint constraint: percent-level retention is sufficient to place the delivered surface reservoir in the terrestrially relevant range.

This apparent pass is conditional on the analytical partition closure. The η -vector is not an SPH-derived measurement, and the calculation does not determine lunar survival, terrestrial spin, total Earth-Moon angular momentum, melt fraction, disk mass, surviving projectile mass, or reaccretion dynamics. Stage 2 therefore identifies a promising water-retention target for the HPC suite rather than a completed collision solution.

A Titan-mass icy body carries so much water that only a few percent need remain near the surface to supply an ocean-scale reservoir. The central impact question is therefore controlled retention, not

whether the projectile contains enough water.

D.3 Stage 3: thermal-stall diagnostic failure

Eight representative grazing cases from the admissible Stage-2 subset were propagated for 100 kyr using 1,000 time steps from an initial temperature of 2800 K. All eight runs exhibited a severe thermal stall: the atmosphere-magma-ocean temperature decreased by only ΔT approximately 0.013 K over the entire 100-kyr interval. This behavior is a numerical artifact of the uncalibrated default effective heat capacity, $C_{\text{eff}} = 5.0 \times 10^{30} \text{ J K}^{-1}$, which is too large for the intended reduced-order magma-ocean/troposphere energy reservoir. The resulting pressure-temperature path therefore fails to enter the ocean-condensation and chemical-quench regime.

The Stage-3 failure is treated as a calibration failure of the present closure, not as evidence that a post-impact Earth could not cool. It also contaminates all downstream modules that depend on the thermal trajectory. Consequently, the escape/isotope and chemistry outputs below are diagnostic responses to the current stalled trajectory and must be rerun after thermal recalibration.

The failed cooling run was scientifically useful because it revealed a numerical bottleneck rather than hiding it downstream. Correcting this one thermal parameter became a decisive test of whether the atmosphere, escape, and chemistry chain could operate at all.

D.4 Stage 4: escape and isotope discrepancy

Under the stalled 100-kyr thermal trajectories, the maximum Hunten crossover mass reached approximately $m_c = 123.7 \text{ amu}$, formally exceeding the masses of atomic nitrogen and ^{36}Ar and therefore permitting their entrainment in the reduced-order escape prescription. Integrated nitrogen loss, however, was only about 0.14%: approximately 99.86% of the nominal nitrogen reservoir remained, corresponding to a final mass of $2.347 \times 10^{21} \text{ kg}$. This falls far short of the order- 1.7×10^{-3} long-term residual fraction required by the NH_3 -rich reference normalization. In the same runs, the toy ^{36}Ar reservoir was effectively stripped entirely, producing an aggressive N/Ar separation.

The exchangeable-water isotope ratio also moved away from the terrestrial target under the initial impactor-water assumption. Final D/H values were $(1.958\text{--}1.985) \times 10^{-4}$, or approximately 1.26–1.27 times SMOW. Within this reduced-order setup, the fixed impactor value therefore cannot be treated as a successful input. R_i must remain a free inverse parameter, and a lower-than-SMOW trial value is indicated for the next calibration run if fractionating escape is to evolve the mixed reservoir toward 1.558×10^{-4} . This is a screening inference, not a measurement of primordial Titan-like water.

The model cannot match nitrogen by simply maximizing escape: the same process can destroy the noble-gas and isotope agreement. This converts three observables into a joint constraint and makes arbitrary one-variable tuning much harder.

D.5 Stage 5: finite-feedstock chemistry grid

A 16-case chemistry grid was evaluated using four impactor ammonia mass fractions, $x_{\text{NH}_3} = 0.005, 0.010, 0.025, \text{ and } 0.050$, crossed with redox states $\Delta \text{IW} = -3, -2, -1, \text{ and } 0$. The finite-reagent network reproduced the expected hydrogen stoichiometric scaling cleanly. The four ammonia levels generated approximately 19.6, 39.1, 97.7, and 195.4 bar H_2 equivalent, respectively. In particular, the 2.5 wt.% NH_3 case independently recovered the approximately 97-bar H_2 normalization used in the main text.

All 16 diagnostic chemistry cases were intentionally aborted before a nominal HCN yield was evaluated. The imported Stage-3 trajectory remained near 2800 K and never entered the declared $T < 1500$ K synthesis/quench window. Accordingly, the executable pipeline reports 0/16 successful cases and no physical HCN yield, rather than manufacturing $\text{HCN}_{\text{final}} = 0$ values outside the network's applicability domain. The chemically meaningful test is the same finite-feedstock grid driven by a physically calibrated cooling trajectory.

The initial absence of HCN traced back to the stalled thermal history, not automatically to impossible prebiotic chemistry. That dependency demonstrates why the modules must be evaluated as a causal chain rather than as isolated calculations.

D.6 Incremental interpretation and next simulation tier

Taken together, the first local screening does not yield a joint viable solution, but neither does it falsify the hypothesis. It establishes one encouraging reduced-order result - a non-empty grazing subset with a percent-level surface-water partition in the nominal terrestrial footprint - while exposing four coupled restrictions: an excessive high-velocity tail in the local delivery surrogate, an uncalibrated thermal reservoir, insufficient integrated nitrogen removal combined with excessive ^{36}Ar stripping, and a downstream isotope/HCN response that is incompatible with the stalled thermal path. The correct role of these calculations is therefore to prune and rank Ω before computationally expensive deployment.

The incremental strategy is to preserve Appendix C unchanged as the submission-grade research program, recalibrate the Stage-3 thermal reservoir first, propagate the resulting pressure-temperature history into Stage 4, and only then rerun Stage 5. The surviving local solutions can subsequently define initial-condition families for the full N-body and SPH calculations. Future runs should be appended to this data log rather than replacing earlier diagnostic results, so that the evolution of the joint parameter space and each calibration decision remains auditable.

The first campaign did what a useful screening model should do: it retained a promising impact family, exposed the failure points, and established the correct order for repairing them. This sharply narrows the work that must be transferred to expensive N-body and SPH calculations.

D.7 Calibration Run 2: thermal recovery and downstream re-evaluation

A second local CPU screening was performed after recalibrating the reduced thermal reservoir, releasing the impactor-water D/H ratio as an inverse parameter, increasing the finite H_2 escape normalization, exposing an atomic-N photolysis/homopause accessibility scale, and coupling Stage 5 explicitly to the cooling Stage-3 trajectory below a 1500 K quench threshold. This run is recorded incrementally and does not supersede the first diagnostic results in D.1-D.6.

Stage 1 was repeated as five independent ensembles of 20,000 objects, again totaling 100,000 samples. The second ensemble produced 173 focused collision vectors, 28 of which lay in the preferred 60° - 75° grazing window. The median impact speed of the unrestricted accepted population was approximately 48.25 km s⁻¹, confirming the persistent high-velocity tail seen in the first screening. Across the four Stage-2 impactor masses, 112 grazing mass/trajectory combinations were generated; imposing $1 \leq v_i/v_{\text{esc}} \leq 2$ retained 24. Within this intended subset the analytical surface-water inventory spanned approximately 1.15-3.34 modern oceans, with a median of approximately 2.01 oceans. The water-footprint result is therefore reproducible across independent Monte Carlo seeds at reduced-order level.

The Stage-3 thermal recalibration eliminated the numerical stall. Eight representative admissible grazing cases were propagated for 100 kyr using 1,000 steps with $C_{\text{eff}} = 2.5 \times 10^{25} \text{ J K}^{-1}$. All eight trajectories cooled monotonically from 2800 K, crossed below 1500 K at approximately 50.55 kyr, and reached the imposed 300 K temperature floor before the end of the integration. The previous Delta T approximately 0.013 K failure is thus confirmed to have been a consequence of the oversized default C_{eff} rather than an intrinsic outcome of the reduced energy-balance architecture. The exact cooling history remains exploratory because radiative transfer, magma-ocean convection, latent heat, volatile dissolution/exsolution, and evolving opacity are not solved self-consistently.

Stage 4 responded strongly to the revised finite H₂ reservoir and atomic-N accessibility parameterization. The final nitrogen mass was $4.322 \times 10^{18} \text{ kg}$ in the representative runs, equivalent to a residual fraction of 1.839×10^{-3} of the $2.35 \times 10^{21} \text{ kg}$ reference reservoir. This lies within about 8.2% of the nominal 1.7×10^{-3} filtering target and is a substantial improvement over the first run's approximately 99.86% retention. However, the same normalization drove the maximum crossover mass to approximately $5.65 \times 10^5 \text{ amu}$ and stripped the toy ³⁶Ar inventory to effectively zero. This is not considered a joint escape/isotope pass: it indicates that the H₂-wind/photolysis closure can force the nitrogen target but presently over-processes noble gases.

The inverse D/H trial also over-corrected. With R_i initialized at 1.35×10^{-4} , final exchangeable-water D/H spanned approximately $1.357\text{--}1.371 \times 10^{-4}$, or about 0.87–0.88 SMOW. The direction of the discrepancy therefore reversed relative to the first screening, demonstrating that R_i is genuinely a sensitive inverse parameter but that 1.35×10^{-4} is not an acceptable final calibration under the current escape history. A subsequent search must solve R_i jointly with water retention and fractionating escape rather than fixing it independently.

Stage 5 was rerun as the same 16-case grid spanning $x_{\text{NH}_3} = 0.005, 0.010, 0.025, \text{ and } 0.050$ and Delta IW = -3 to 0, now using the actively cooling pressure-temperature trajectory. The chemistry module entered the $T < 1500 \text{ K}$ synthesis/quench window at approximately 50.55 kyr. Surviving HCN became non-zero in all 16 cases, with final inventories of approximately 8.793×10^{16} to $1.736 \times 10^{17} \text{ mol}$ across the grid. The H₂ pressure normalizations remained approximately 19.6, 39.1, 97.7, and 195.4 bar for the four NH₃ abundance levels. The result remains a reduced-network output, not a laboratory-validated impact yield.

Calibration Run 2 therefore converts one major numerical failure into a working coupled trajectory but exposes the next bottleneck clearly. Thermal-quench coupling now behaves consistently and nitrogen filtering can be driven near the reference mass target, yet the same escape closure catastrophically over-strips ³⁶Ar and the current low- R_i trial undershoots SMOW. The next local calibration should target the intersection of nitrogen, D/H, and noble-gas survival rather than further tuning water or H₂ stoichiometry. Only parameter combinations that improve these observables jointly should be promoted to the full HPC initial-condition set.

Once the thermal stall was repaired, the framework produced cooling, nitrogen loss, and HCN survival in one connected calculation. The remaining isotope and argon failures are therefore specific, measurable targets rather than a general collapse of the hypothesis.

D.8 Reproducibility Run 3: fixed-parameter verification of the calibrated chain

A third local execution was performed strictly as a reproducibility test of Calibration Run 2. No additional calibration parameters were changed. The same eight preserved Stage-2 representative grazing inputs were propagated through the patched 100-kyr Stage-3/4 solver using 1,000 time steps,

after which the complete 4 x 4 Stage-5 chemistry grid was rerun against the representative Titan-mass cooling trajectory. The purpose was to distinguish a reproducible numerical result from a one-off execution artifact.

All eight thermal trajectories again cooled monotonically from 2800 K, crossed below 1500 K at 50.5506 kyr, and reached the imposed 300 K floor. The escape module again returned a final nitrogen mass of 4.322×10^{18} kg, corresponding to a residual fraction of 1.83935×10^{-3} , approximately 8.2% above the nominal 1.7×10^{-3} target. The ^{36}Ar reservoir was again driven effectively to zero. Final exchangeable D/H across the eight cases ranged from 1.3574×10^{-4} to 1.3705×10^{-4} , equivalent to approximately 0.871-0.880 SMOW. Thus the favorable nitrogen filtering and the unfavorable noble-gas and isotope outcomes are reproducible properties of the present reduced-order calibration rather than numerical noise.

The full 16-case chemistry grid reproduced non-zero surviving HCN in every case. HCN_{final} ranged from 8.793×10^{16} to 1.736×10^{17} mol across $x_{\text{NH}_3} = 0.005$ -0.050 and $\Delta \text{IW} = -3$ to 0, with first entry into the $T < 1500$ K synthesis/quench domain again occurring at 50.5506 kyr. This fixed-parameter replay demonstrates internal numerical reproducibility; it does not establish a complete cyanosulfidic pathway, global HCN abundance, or laboratory-validated impact yield.

Reproducibility Run 3 therefore strengthens confidence in the numerical implementation while leaving the scientific bottleneck unchanged. The next calibration target is not further enhancement of HCN or nitrogen loss, but the joint intersection of nitrogen retention, D/H, and noble-gas survival under a physically richer upper-atmosphere and escape treatment. These outputs should continue to be used as screening constraints on Omega rather than as substitutes for the submission-grade calculations specified in Appendix C.

The same outcomes reappeared without changing the parameters. That makes the successful and unsuccessful outputs properties of the implemented model rather than one-off numerical accidents, which is the minimum foundation needed for meaningful sensitivity analysis.

D.9 Calibration Run 4: isotope-bracket refinement and corrected noble-gas exposure semantics

Run 4 modified only the Stage-4 boundary parameters while preserving the verified thermal and nitrogen architecture. The impactor-water isotope ratio was set to $R_i = 1.465 \times 10^{-4}$. The historical coefficient 0.042 represented the exposed argon fraction; under the corrected executable semantics it is therefore replayed as $\text{argon_shielding_factor} = 0.958$ and $\text{argon_exposure_factor} = 0.042$. C_{eff} remained fixed at $2.5 \times 10^{25} \text{ J K}^{-1}$, and the same eight preserved atmosphere cases were propagated for direct comparison with the reproducibility baseline.

The thermal solution was unchanged: all eight trajectories again crossed below 1500 K at 50,550.55 yr and cooled to the imposed 300 K floor. Nitrogen filtering also remained stable, with $M_{\text{N,final}} = 4.32247 \times 10^{18}$ kg and a residual fraction of 1.83935×10^{-3} , approximately 8.2% above the nominal 1.7×10^{-3} target. The Stage-5 stoichiometric normalization remained approximately 97.7388 bar H₂ for $x_{\text{NH}_3} = 0.025$, and HCN survival remained positive in 16/16 chemistry cases.

The isotope adjustment improved the D/H trajectory and placed it within a near-SMOW lower-side envelope. Across the eight atmosphere cases, final exchangeable D/H ranged from 1.46837×10^{-4} to 1.47434×10^{-4} , equivalent to approximately 0.9425-0.9463 SMOW. This does not establish exact isotopic closure, but the difference is not resolved meaningfully by the present reduced-order treatment given

the uncertainties in impactor-water D/H, escape fractionation, mantle exchange, and the pre-impact terrestrial reservoir. No additional late-veneer term is therefore imposed in the present calibration. Later accretion may be examined within a complete water and isotope mass-balance calculation, but it is not required as a fitted component of this model.

The Run-4 noble-gas boundary trial remained quantitatively insufficient. Starting from the model's primitive ^{36}Ar reference inventory of 5.0×10^{15} kg, 5.1137×10^{13} kg survived, corresponding to 1.022745% retention. Thus an exposure fraction of 0.042 (shielding fraction 0.958) does not reach the model's 12% calibration target under the present ~ 97 -bar H_2 escape normalization. This does not identify a physical protection mechanism; homopause diffusion resistance, delayed upper-atmosphere access, transient mantle sequestration, later outgassing, and revised escape physics remain hypotheses to test.

Run 4 moved D/H close enough to define a concrete mixing calculation, while the argon miss exposed the next hard physical constraint. In practical terms, the framework was becoming more predictive because each improvement left a smaller and more clearly defined problem.

D.10 Calibration Runs 5-6 and Runs 7-9 computational extensions: local calibration, robustness, N-body validation, N-body campaign readiness, and SPH preparation

Run 6 extends the Run-5 calibrated state into an explicit one-at-a-time (OAT) response-surface and counterfactual audit. No sweep point was recalibrated after perturbation. The complete evidence is stored under `results_cached/appendix_d/d10_calibration_extensions/run6_robustness/`; no earlier archive is required.

D.10.1 Reproducibility and mechanistic verification

- **Run-5 baseline reproduction** — PASS with zero numerical regression. Run 6 reproduces the preserved Run-5 atmosphere baseline: $T < 1500$ K at 50,550.55 yr in all eight cases; final D/H = 0.942470–0.946301 SMOW; nitrogen residual = 1.8393505×10^{-3} ; ^{36}Ar retained fraction = 0.120000001; and maximum crossover mass = 5.64603×10^5 amu. The machine-readable baseline and sweep definitions are in `results_cached/appendix_d/d10_calibration_extensions/run6_robustness/run_summary.json`; aggregate responses are in `aggregate_results.csv` and per-case values in `case_results.csv`.

- **Hydrogen-drag ablation** — **MECHANISTIC NECESSITY demonstrated within the surrogate**. Setting the H_2 escape reservoir to zero leaves both nitrogen and ^{36}Ar at 100% of their initialized inventories; the maximum crossover mass collapses from 5.64603×10^5 amu to 2.016 amu. Thus the selective-loss behavior in this implementation is generated by the hydrogen-drag engine rather than by an unrelated bookkeeping term. This is a model-internal mechanistic ablation, not independent empirical validation of the early-Earth H_2 reservoir.

- **Atomic-nitrogen accessibility ablation** — **NECESSARY FOR N FILTERING**. With `n_photolysis_scale = 0`, the nitrogen residual is 1.0 while the calibrated argon response is otherwise preserved. The result isolates atomic-N accessibility/photolysis as a necessary modeled pathway for nitrogen loss under the present closure.

- **Argon shielding endpoints** — **EXPECTED LIMITS RECOVERED**. With no argon shielding, the retained ^{36}Ar fraction falls to approximately 2.27×10^{-53} ; with complete shielding it is 1.0. These endpoint ablations verify the implemented exposure semantics introduced in Run 5.

The calibrated result is not merely a collection of fitted numbers: removing the proposed mechanisms makes the corresponding effects disappear, and the limiting cases behave correctly. That mechanistic response is stronger evidence for a functioning numerical architecture than target matching alone.

D.10.2 Sensitivity analysis and calibration limits

To quantify the structural coupling of the code, the volatile diagnostics under one-at-a-time (OAT) parameter variations are mapped in Table D1.

Table D1: One-at-a-Time (OAT) Response Surface and Model Sensitivity Matrix

Parameter / Sweep Case	Perturbation Scale	Nitrogen Residual (N _{final} / N ₀)	Retained ³⁶ Ar Fraction	Max Crossover Mass (amu)	Final D/H Range (SMOW)
Run-5 / Run-6 Baseline	1.0×	1.8393505 × 10⁻³	0.120000001 (12.0000001%)	564,603	0.942470–0.946301
<i>³⁶Ar Exposure Sweeps</i>	0.5×	1.8393505 × 10 ⁻³	0.3468 (34.68%)	564,603	0.942470–0.946301
	0.8×	1.8393505 × 10 ⁻³	0.1835 (18.35%)	564,603	0.942470–0.946301
	1.2×	1.8393505 × 10 ⁻³	0.07845 (7.845%)	564,603	0.942470–0.946301
	1.5×	1.8393505 × 10 ⁻³	0.04144 (4.144%)	564,603	0.942470–0.946301
<i>Nitrogen Photolysis Sweeps</i>	0.8×	6.3907 × 10 ⁻²	0.120000001	564,603	0.942470–0.946301
	0.9×	1.5124 × 10 ⁻²	0.120000001	564,603	0.942470–0.946301
	1.1×	2.7066 × 10 ⁻⁴	0.120000001	564,603	0.942470–0.946301
	1.2×	5.2727 × 10 ⁻⁵	0.120000001	564,603	0.942470–0.946301
<i>H₂ Reservoir Sweeps</i>	0.8×	2.4172 × 10 ⁻²	~0.12	451,683	0.942470–0.946301
	1.2×	4.6009 × 10 ⁻⁴	~0.12	677,523	0.942470–0.946301
<i>Isotope Target Sweeps</i>	-5% R _i	1.8393505 × 10 ⁻³	0.120000001	564,603	0.8971–0.9039
	-2% R _i	1.8393505 × 10 ⁻³	0.120000001	564,603	0.9243–0.9293
	+2% R _i	1.8393505 × 10 ⁻³	0.120000001	564,603	0.9606–0.9633
	+5% R _i	1.8393505 × 10 ⁻³	0.120000001	564,603	0.9878–0.9887
	SMOW Input	1.8393505 × 10 ⁻³	0.120000001	564,603	1.00006–1.00019

- **³⁶Ar exposure response — CALIBRATION-SENSITIVE.** Around the Run-5 exposure fraction 0.0194752434, OAT factors of 0.5, 0.8, 1.0, 1.2 and 1.5 produce retained ³⁶Ar fractions of 0.3468, 0.1835, 0.1200, 0.07845 and 0.04144, respectively. The 12% value is therefore a calibrated intersection on a steep response surface, not an independent prediction. The scientific information supplied by this axis is the exposure required by the surrogate to satisfy the selected argon bookkeeping target.

- **Nitrogen photolysis response — STRONGLY CALIBRATION-SENSITIVE.** Varying n_{photolysis_scale} from 0.8× to 1.2× its baseline value changes the nitrogen residual from 6.3907 × 10⁻² through 1.5124 × 10⁻², 1.83935 × 10⁻³, 2.7066 × 10⁻⁴ and 5.2727 × 10⁻⁵. This axis must therefore be treated as a constrained surrogate parameter until independently derived photochemical accessibility is available.

- **H₂-reservoir response — STRONGLY COUPLED TO N LOSS.** Scaling the H₂ reservoir from 0.8× to 1.2× changes the nitrogen residual from 2.4172 × 10⁻² to 4.6009 × 10⁻⁴, while the calibrated ³⁶Ar retained fraction remains near 0.12 because its exposure coefficient is independently specified. The

response demonstrates both the physical role of hydrogen drag in the surrogate and the need for an externally constrained H₂ inventory.

• **D/H response to R_i — FREE-PARAMETER CONTROL CONFIRMED.** Perturbing impactor-water D/H by -5%, -2%, baseline, +2% and +5% produces final envelopes of approximately 0.8971–0.9039, 0.9243–0.9293, 0.9425–0.9463, 0.9606–0.9633 and 0.9878–0.9887 SMOW. A SMOW-like impactor-water counterfactual returns approximately 1.00006–1.00019 SMOW. The isotope result is therefore explicitly conditional on R_i, as required by the open-mixing formulation, and is not counted as an independent prediction.

• **Thermal-capacity response — ONE-SIDED ROBUSTNESS WITH AN EXPLICIT FAILURE BOUNDARY.** For C_{eff} = 2.50, 2.75, 3.125 and 3.75 × 10²⁵ J K⁻¹, the atmosphere crosses below 1500 K at approximately 50.55, 55.56, 63.16 and 75.78 kyr. At C_{eff} = 5.00 × 10²⁵ J K⁻¹, it remains at 1512.29 K after 100 kyr and never enters the modeled quench window. Because the Run-5 baseline is already at the lower bound of the declared C_{eff} interval, this robustness test is necessarily one-sided.

The sensitivity sweep shows which conclusions survive reasonable changes and which depend steeply on calibration. This gives reviewers an explicit map of the model's strong, weak, and falsifiable regions instead of presenting one favorable parameter set in isolation.

D.10.3 Chemical robustness and falsification boundary

The effective thermal capacity (C_{eff}) sets the pace of post-impact atmospheric relaxation. Table D2 maps the coupled response of the downstream quench-chemistry network across alternative thermal paths.

Table D2: Thermal Capacity Scaling and Downstream Prebiotic Chemistry Yields

Effective Heat Capacity (C _{eff} , J K ⁻¹)	Cooling Timeline (T < 1500 K)	4×4 Chemistry Grid Success Rate	HCN Abundance Range (mol)	Evaluation Status
2.50 × 10 ²⁵ (Baseline)	50.55 kyr	16 / 16 positive cases	8.793 × 10 ¹⁶ –1.736 × 10 ¹⁷	Valid baseline intersection
2.75 × 10 ²⁵	55.56 kyr	16 / 16 positive cases	8.793 × 10 ¹⁶ –1.736 × 10 ¹⁷	Thermally robust
3.125 × 10 ²⁵	63.16 kyr	16 / 16 positive cases	8.793 × 10 ¹⁶ –1.736 × 10 ¹⁷	Thermally robust
3.75 × 10 ²⁵	75.78 kyr	16 / 16 positive cases	8.793 × 10 ¹⁶ –1.736 × 10 ¹⁷	Thermally robust
5.00 × 10 ²⁵	Stalled (>100 kyr)	0 / 16 (aborted by design)	Not evaluated	Falsification boundary

• **HCN chemistry propagation — ROBUST CONDITIONAL RESULT.** The complete 4 × 4 chemistry grid was propagated through each thermal trajectory that entered the T < 1500 K quench window. At C_{eff} = 2.50, 2.75, 3.125 and 3.75 × 10²⁵ J K⁻¹, all 16/16 chemistry cases remain HCN-positive, for 64/64 successful propagated cases overall. HCN_{final} remains approximately 8.793 × 10¹⁶–1.736 × 10¹⁷ mol. Generated summaries and raw CSV files are stored below [results_cached/appendix_d/d10_calibration_extensions/run6_robustness/](#).

• **Chemistry failure boundary — PASS AS FALSIFICATION BEHAVIOR.** At C_{eff} = 5.00 × 10²⁵ J K⁻¹, the atmosphere does not cross below 1500 K within 100 kyr; consequently all 16 chemistry cases are intentionally aborted rather than allowed to generate a nominal HCN result outside the model's thermal applicability window. The Run-6 chemistry layer therefore does not produce HCN unconditionally: successful chemistry is contingent on entry into the specified cooling/quench regime.

- **Interpretation of the HCN result.** The 64/64 result demonstrates robustness of HCN positivity to the tested thermal-timing perturbations within the finite reduced network. It does not establish a laboratory-validated HCN yield, a complete cyanosulfidic pathway, or historical realization on the Hadean Earth. Those remain submission-grade chemistry requirements.

HCN remains positive across every tested thermal history that actually enters the declared quench window, while the workflow suppresses nominal HCN output outside that domain. The result is therefore conditional but non-trivial: the chemistry is robust within its stated applicability range and fails transparently beyond it.

D.10.4 Evidential classification after Run 6

- **Mechanistic windows — STRENGTHENED.** Run 6 shows that hydrogen escape is necessary for the modeled N/Ar filtering; atomic-N accessibility is necessary for modeled N loss; the argon exposure semantics recover the correct limiting behavior; and HCN positivity survives substantial changes in thermal timing provided the quench window is reached.

- **Calibration-sensitive quantities — EXPLICITLY IDENTIFIED.** The 12% ^{36}Ar retention, the nominal nitrogen residual, and the final D/H bracket are not reclassified as independent predictions. Run 6 quantifies their dependence on argon exposure, $n_{\text{photolysis_scale}}/\text{H}_2$ inventory, and R_i , respectively.

- **Audit-ready response surface — PASS at local reduced-order level.** Run 6 converts the previous single calibrated point into explicit OAT response surfaces and ablations. It therefore reduces the risk of presenting a post-hoc calibrated intersection as a broad or unique physical solution. No new empirical tolerance was invented to force the perturbed cases into PASS/FAIL bins.

- **Canonical codebase and results.** All executable Appendix D results are regenerated from the single distributed codebase at GitHub. Run 6 is stored under `results_cached/appendix_d/d10_calibration_extensions/run6_robustness/`, including summary CSV/JSON, case results, chemistry propagation, and raw outputs. Historical runs are replayed by configuration from this same codebase at GitHub.

- **Remaining validation layers — OPEN.** Run 6 does not replace submission-grade N-body delivery statistics, SPH Earth–Moon/volatile-partition calculations, resolved atmosphere–magma-ocean and multi-species escape modeling, independent noble-gas and nitrogen-isotope predictions, or experimentally/kinetically constrained shock and prebiotic chemistry. These remain the principal external-validation program.

- **Current evidential status after Run 6.** The reduced-order framework now supports a narrower and stronger statement than a simple successful calibration: it contains an exactly reproducible calibrated intersection; exposes steep calibration axes rather than concealing them; demonstrates internal mechanistic necessity through ablations; identifies a thermal falsification boundary; and preserves HCN positivity across all 64 chemistry cases for which the modeled atmosphere actually reaches the prescribed quench window. It does not establish uniqueness, dynamical probability, historical occurrence, or submission-grade physical validation.

Run 6 turns a calibrated point into an auditable response surface. It strengthens the paper’s central claim from ‘one setup can be made to work’ to ‘the code reveals which mechanisms, parameters, and failure boundaries control the provisional solution.’

D.10.5 Run 7: executable N-body path and SPH campaign definition

Run 7 adds two isolated extension modules within the same final codebase. The historical `delivery_monte_carlo.py` surrogate remains available for the Appendix D replay, while `nbody/` supplies a genuine REBOUND integration path and `sph/` supplies SWIFT-oriented campaign preparation and snapshot diagnostics. Generated evidence is stored under `results_cached/appendix_d/d10_calibration_extensions/run7_nbody/` and `results_cached/appendix_d/d10_calibration_extensions/run7_sph/`. The archive label records software versioning only; it does not promote the dynamical or hydrodynamic components to calibrated physical evidence.

N-body collision-logger validation result

A deterministic IAS15 test integrated 32 massless validation particles for 0.20 yr and recovered all 32 intended Earth collisions. Earth's collision radius was deliberately inflated to 50 R_E , so this run validates integration, event removal, collision serialization, encounter geometry, and energy bookkeeping only. The recorded numerical results are given in Table D3.

Table D3: Run-7 REBOUND Collision-Logger Validation

Recorded metric	Run-7 value
Evidence class	Pipeline validation
REBOUND version / integrator	5.1.1 / IAS15
Collision search	Line
Seed / integration duration	410006 / 0.20 yr
Initial test particles	32
Earth-radius multiplier	50.0 (synthetic validation only)
Earth collisions logged	32 / 32
Impacts flagged at 60°-75°	0 / 32
Relative energy error	-2.2860×10^{-16}

The zero count in the 60°-75° column is not a physical rejection of the grazing window: this synthetic generator was designed to force collision events across a broad impact-parameter range, not to sample or optimize the Moon-survival geometry. Its scientific output is the successful event-logging test and near-machine-precision energy closure.

Physical-radius outer-reservoir control result

A second REBOUND run used the present Sun-Earth-Jupiter-Saturn-Uranus-Neptune architecture, 256 massless particles initialized from 15-45 au, MERCURIUS with native collision search, and Earth's physical radius. Over the declared 200-yr control interval it recorded zero Earth collisions. Table D4 reports the complete run-level result.

Table D4: Run-7 Physical-Radius Outer-Reservoir Control

Recorded metric	Run-7 value
Evidence class	Exploratory

Recorded metric	Run-7 value
REBOUND version / integrator	5.1.1 / MERCURIUS
Collision search / time step	Native particle-overlap search / 0.02 yr
Seed / integration duration	410100 / 200 yr
Initial test particles	256
Source semimajor-axis interval	15-45 au
Earth-radius multiplier	1.0 (physical radius)
Earth collisions logged	0 / 256
Relative energy error	-5.6098×10^{-9}

The zero-collision outcome is an expected null control for a short integration of the present planetary architecture. It is not a Nice-model instability, contains no delayed-release mechanism, and is many orders of magnitude too short to estimate a ~ 4.1 -Ga source-to-Earth delivery probability. Run 7 therefore establishes an executable and auditable N-body path, not a dynamical feasibility pass.

SPH campaign-definition result

The SWIFT-oriented preparation layer generated a deterministic Cartesian matrix over impactor mass, mutual-escape-speed ratio, impact angle, water fraction, and particle count. The matrix contains 96 unique run identifiers. Derived contact geometry and velocity ranges were recalculated from the generated JSON and are summarized in Table D5.

Table D5: Run-7 SWIFT/SPH Campaign Matrix

Parameter axis or derived range	Declared values	Levels / cases
Impactor mass	0.75, 1.00, 1.25 M _{Titan}	3
Speed ratio	1.0, 1.2 v _{esc,mut}	2
Impact angle	60°, 65°, 70°, 75°	4
Impactor water mass fraction	0.30, 0.50	2
Particle count	100,000; 1,000,000	2
Cartesian matrix	$3 \times 2 \times 4 \times 2 \times 2$	96 cases
100,000-particle class	Pipeline validation	48 cases
1,000,000-particle class	HPC resolution study	48 cases
Mutual escape speed	9.467-9.647 km s ⁻¹	Derived range
Impact speed	9.467-11.576 km s ⁻¹	Derived range
Contact impact parameter	b/R = 0.8660-0.9659	Derived range

No SPH impact is reported in Run 7. The 100,000-particle entries are pipeline tests, while the 1,000,000-particle entries begin a future resolution study and are not automatically converged or publication-grade. Hydrostatically relaxed differentiated bodies, planetary equations of state, cluster-specific SWIFT configurations, executed snapshots, resolution convergence, remnant/disk classification, volatile partitioning, and coupled Earth-Moon orbital outcomes remain open.

Run 7 establishes that the N-body event pipeline can record known collisions accurately, that a short physical-radius control produces the expected null result, and that the future SPH campaign has a complete, reproducible experiment matrix. The computational route is therefore operational even though the decisive physical simulations remain unexecuted.

D.10.6 Evidential classification after Run 7

Run 7 closes an implementation gap rather than a physical-validation gap. A real gravitational integrator now replaces neither the historical surrogate record nor the need for a literature-anchored, multi-seed instability ensemble. Likewise, the SPH matrix and HDF5 analysis utilities define a reproducible HPC interface but do not supply hydrodynamic outcomes. The submission-grade N-body delivery probability and convergence-tested SPH Earth-Moon/volatile-partition calculations therefore remain open criteria in Appendix C.

This distinction strengthens the paper: the proposed tests are no longer only a wish list, but the manuscript does not mistake software readiness for proof of a 4.1-Ga event. Reviewers can see exactly what is implemented and exactly what remains open.

D.10.7 Run 8: multi-seed post-release N-body transport pilot

Run 8 extends `nbody/` with a deterministic ensemble driver and a perihelion-injection generator, executed from the same final codebase. Four IAS15 seeds (410201-410204) each integrated 256 massless post-release tracers for 500 yr using line collision search and Earth's physical radius. The injected distribution was conditioned on $a = 15\text{-}30$ au, $q = 0.75\text{-}1.75$ au, and a Rayleigh inclination scale of 10° . It is a post-release transport kernel, not a self-consistent instability trigger. Outputs are stored under `results_cached/appendix_d/d10_calibration_extensions/run8_nbody/`.

Post-release transport result

The ensemble integrated 1,024 trajectories, corresponding to 512,000 particle-years, and recorded zero physical-radius Earth collisions. The two-sided 95% Wilson interval therefore has an upper bound of 3.7374×10^{-3} collisions per integrated particle for this declared 500-yr conditioned experiment. No $60^\circ\text{-}75^\circ$ collision geometry was available because no Earth collision occurred. The maximum absolute run-level relative energy error was 1.9630×10^{-15} . Table D6 gives the complete aggregate result.

Table D6: Run-8 Multi-Seed Post-Release N-body Transport Pilot

Recorded metric	Run-8 value
Evidence class	Exploratory
Transport stage	Post-release injection kernel
REBOUND version / integrator	5.1.1 / IAS15
Collision search / Earth radius	Line / physical (1.0 R_E)
Seeds	410201, 410202, 410203, 410204
Duration / particles per seed	500 yr / 256
Injected a / q intervals	15-30 au / 0.75-1.75 au
Inclination distribution	Rayleigh, $\sigma = 10^\circ$

Recorded metric	Run-8 value
Total trajectories / particle-years	1,024 / 512,000
Physical-radius Earth collisions	0 / 1,024
Impacts flagged at 60°-75°	0 (no collisions recorded)
95% Wilson upper bound	3.7374×10^{-3} per integrated particle
Maximum relative energy error	1.9630×10^{-15}

This null result neither supports nor rejects the proposed Titan-mass event. The experiment begins after an assumed release onto high-eccentricity orbits, uses massless tracers, samples only 500 yr per seed, and contains too few trajectories to resolve a rare physical collision channel or its grazing-angle distribution. Its contribution is narrower: it demonstrates reproducible multi-seed orchestration, physical-radius collision handling, aggregate provenance, and near-machine-precision energy closure for the post-release transport layer.

Zero collisions are informative here because they prevent a short, conditioned pilot from being mistaken for an easy-delivery demonstration. The verified multi-seed kernel is now ready to be scaled and coupled to self-consistent instability histories, where a physically meaningful delivery probability can actually be tested.

D.10.8 Run 9: literature-anchored N-body campaign readiness

The Run-9 source implementation is stored under `nbody/instability_ensemble/` in the code archive. As part of the controlled D.10 replay, its generated evidence is stored under `results_cached/appendix_d/d10_calibration_extensions/run9_nbody/`: campaign construction and validation records are under `campaign_validation/`, while the executed two-seed smoke-test records are under `local_smoke/`.

The implementation contains a five-giant-planet period-ratio template and a 20-Earth-mass outer planetesimal disk anchored to the five-planet instability experiments of Nesvorný and Morbidelli (2012) [44]. The 1,000 production-candidate disk particles carry mass and back-react on the planets. Deterministic campaign validation passed for the nominal 3:2-3:2-2:1-3:2 period-ratio template, represented disk mass, disk-planet separation, and enabled back-reaction. The declared production campaign contains eight seeds and 10 Myr per seed, but it has not been executed and therefore supplies no delivery probability.

Table D7: Run-9 Literature-Anchored N-body Campaign Readiness

Recorded metric	Run-9 value
Evidence class	Campaign preparation + pipeline validation
Outer-planet template	Five giants; 3:2-3:2-2:1-3:2 nominal period ratios
Outer disk	20 M_Earth; 1,000 back-reacting particles in production config
Declared production campaign	8 seeds × 10 Myr; not executed

Recorded metric	Run-9 value
Campaign construction validation	PASS
Local smoke test	2 seeds; 32 disk particles per seed; 2 yr per seed
Local smoke exposure	64 particles; 128 particle-years
Earth collisions in smoke test	0; not a delivery-probability constraint
Maximum relative energy error	8.0338×10^{-9}

A separate two-seed, two-year local smoke test integrated 64 deliberately coarse disk particles for 128 particle-years, recorded zero Earth collisions, and reached a maximum absolute relative energy error of 8.0338×10^{-9} . These values validate only the new resonant-chain/back-reaction software path. They do not measure an instability timescale or a collision probability. The configuration also does not implement the hydrodynamic gas-disk rebound torques of Liu, Raymond and Jacobson (2022). Submission-grade closure still requires the long N-body campaign, convergence and Earth-Moon encounter refinement; the 96-case SWIFT/SPH execution under sph/production_campaign/ remains separate HPC work.

Research Note

This revision intentionally distinguishes literature-backed boundary conditions from assumptions specific to the proposed impact. Several numerical values in the first draft were corrected by direct dimensional recalculation. Version 4 further anchors the dynamical source classes in giant-planet-instability literature, makes H₂-driven hydrodynamic escape and crossover-mass physics an explicit nitrogen/noble-gas constraint, introduces Titan's measured primordial-argon depletion as a compositional discriminator, and connects impact redox/HCN production directly to cyanosulfidic prebiotic chemistry. These additions strengthen the hypothesis by adding externally testable couplings while preserving the distinction between plausibility and historical evidence. Before journal submission, the dynamical/impact sections still require submission-grade self-consistent instability N-body ensembles and executed, convergence-tested SPH results and the bibliography should be checked against final publisher metadata.

References

1. Bottke, W. F. & Norman, M. D. (2017). The Late Heavy Bombardment. *Annual Review of Earth and Planetary Sciences* 45, 619–647.
2. Morbidelli, A., Nesvorný, D., Laurenz, V., Marchi, S., Rubie, D. C., Elkins-Tanton, L., Wieczorek, M. & Jacobson, S. (2018). The timeline of the lunar bombardment: Revisited. *Icarus* 305, 262–276.
3. Boehnke, P. & Harrison, T. M. (2016). Illusory Late Heavy Bombardments. *Proceedings of the National Academy of Sciences* 113, 10802–10806.
4. Canup, R. M. (2004). Simulations of a late lunar-forming impact. *Icarus* 168, 433–456.
5. Lock, S. J. & Stewart, S. T. (2017). The structure of terrestrial bodies: Impact heating, corotation limits, and synestias. *Journal of Geophysical Research: Planets* 122, 950–982.
6. Waite, J. H. Jr. et al. (2009). Liquid water on Enceladus from observations of ammonia and ^{40}Ar in the plume. *Nature* 460, 487–490.
7. Niemann, H. B. et al. (2010). The composition of Titan's lower atmosphere and simple surface volatiles as measured by the Cassini–Huygens probe gas chromatograph mass spectrometer experiment. *Journal of Geophysical Research: Planets* 115, E12006.
8. Thelen, A. E. et al. (2019). Measurement of CH_3D on Titan at submillimeter wavelengths. *The Astronomical Journal* 157, 219.
9. Nasralla, M. et al. (2025). Solution structure of Titan-relevant aqueous ammonia by neutron diffraction. *Communications Chemistry* 8, 227.
10. Hayes, A. G., Lorenz, R. D. & Lunine, J. I. (2018). A post-Cassini view of Titan's methane-based hydrologic cycle. *Nature Geoscience* 11, 306–313.
11. Guarguaglini, M. et al. (2019). Laser-driven shock compression of 'synthetic planetary mixtures' of water, ethanol and ammonia. *Scientific Reports* 9, 10155.
12. Katyal, N. et al. (2019). Evolution and spectral response of a steam atmosphere for early Earth with a coupled climate–interior model. *The Astrophysical Journal* 875, 31.
13. Nikolaou, A. et al. (2019). What factors affect the duration and outgassing of the terrestrial magma ocean? *The Astrophysical Journal* 875, 11.
14. Goldblatt, C., Robinson, T. D., Zahnle, K. J. & Crisp, D. (2013). Low simulated radiation limit for runaway greenhouse climates. *Nature Geoscience* 6, 661–667.
15. Hartogh, P. et al. (2011). Ocean-like water in the Jupiter-family comet 103P/Hartley 2. *Nature* 478, 218–220.
16. Shimamura, K. et al. (2016). Meteorite impact-induced rapid NH_3 production on early Earth. *Scientific Reports* 6, 38953.
17. Yalinewich, A. & Schlichting, H. E. (2019). Atmospheric mass loss from high-velocity giant impacts. *Monthly Notices of the Royal Astronomical Society* 486, 2780–2789.
18. Biersteker, J. B. & Schlichting, H. E. (2019). Atmospheric mass loss due to giant impacts: the importance of the thermal component for hydrogen–helium envelopes. *Monthly Notices of the Royal Astronomical Society* 485, 4454–4463.

19. Roche, M. J., Lock, S. J., Dou, J., Carter, P. J. & Leinhardt, Z. M. (2025). Atmospheric loss during giant impacts: mechanisms and scaling of near- and far-field loss. arXiv:2505.04343 (preprint; use with appropriate caution).
20. Marty, B. (2012). The origins and concentrations of water, carbon, nitrogen and noble gases on Earth. *Earth and Planetary Science Letters* 313–314, 56–66.
21. Sleep, N. H., Zahnle, K. J., Kasting, J. F. & Morowitz, H. J. (1989). Annihilation of ecosystems by large asteroid impacts on the early Earth. *Nature* 342, 139–142.
22. Abramov, O. & Mojzsis, S. J. (2009). Microbial habitability of the Hadean Earth during the late heavy bombardment. *Nature* 459, 419–422.
23. Bell, E. A., Boehnke, P., Harrison, T. M. & Mao, W. L. (2015). Potentially biogenic carbon preserved in a 4.1 billion-year-old zircon. *Proceedings of the National Academy of Sciences* 112, 14518–14521.
24. Westall, F. et al. (2025). What the earliest evidence for life tells us about the early evolution of the biosphere. *Philosophical Transactions of the Royal Society B* (review; consult final bibliographic metadata before submission).
25. Miller, S. L. (1953). A production of amino acids under possible primitive Earth conditions. *Science* 117, 528–529.
26. Powner, M. W., Gerland, B. & Sutherland, J. D. (2009). Synthesis of activated pyrimidine ribonucleotides in prebiotically plausible conditions. *Nature* 459, 239–242.
27. Patel, B. H., Percivalle, C., Ritson, D. J., Duffy, C. D. & Sutherland, J. D. (2015). Common origins of RNA, protein and lipid precursors in a cyanosulfidic protometabolism. *Nature Chemistry* 7, 301–307.
28. Postberg, F. et al. (2009). Sodium salts in E-ring ice grains from an ocean below the surface of Enceladus. *Nature* 459, 1098–1101.
29. Spilker, L. (2019). Cassini-Huygens' exploration of the Saturn system: 13 years of discovery. *Science* 364, 1046–1051.
30. Hartmann, W. K. (2019). History of the terminal cataclysm paradigm: Epistemology of a planetary bombardment that never (?) happened. *Geosciences* 9, 285.
31. Genda, H. & Abe, Y. (2003). Survival of a proto-atmosphere through the stage of giant impacts: The mechanical aspects. *Icarus* 164, 149–162. doi:10.1016/S0019-1035(03)00101-5.
32. Genda, H. & Abe, Y. (2005). Enhanced atmospheric loss on protoplanets at the giant impact phase in the presence of oceans. *Nature* 433, 842–844. doi:10.1038/nature03360.
33. Fegley, B. Jr., Prinn, R. G., Hartman, H. & Watkins, G. H. (1986). Chemical effects of large impacts on the Earth's primitive atmosphere. *Nature* 319, 305–308. doi:10.1038/319305a0.
34. Maher, K. A. & Stevenson, D. J. (1988). Impact frustration of the origin of life. *Nature* 331, 612–614. doi:10.1038/331612a0.
35. Zahnle, K., Schaefer, L. & Fegley, B. (2010). Earth's earliest atmospheres. *Cold Spring Harbor Perspectives in Biology* 2, a004895. doi:10.1101/cshperspect.a004895.
36. Liu, B., Raymond, S. N. & Jacobson, S. A. (2022). Early Solar System instability triggered by dispersal of the gaseous disk. *Nature* 604, 643–646. doi:10.1038/s41586-022-04535-1.

37. Levison, H. F., Bottke, W. F., Gounelle, M., Morbidelli, A., Nesvorný, D. & Tsiganis, K. (2009). Contamination of the asteroid belt by primordial trans-Neptunian objects. *Nature* 460, 364–366. doi:10.1038/nature08094.
38. Deienno, R., Nesvorný, D., Vokrouhlický, D. & Yokoyama, T. (2014). Orbital perturbations of the Galilean satellites during planetary encounters. *The Astronomical Journal* 148, 25. doi:10.1088/0004-6256/148/2/25.
39. Hunten, D. M., Pepin, R. O. & Walker, J. C. G. (1987). Mass fractionation in hydrodynamic escape. *Icarus* 69, 532–549.
40. Gronoff, G. et al. (2020). Atmospheric escape processes and planetary atmospheric evolution. *Journal of Geophysical Research: Space Physics* 125, e2019JA027639. doi:10.1029/2019JA027639.
41. Niemann, H. B. et al. (2005). The abundances of constituents of Titan’s atmosphere from the GCMS instrument on the Huygens probe. *Nature* 438, 779–784. doi:10.1038/nature04122.
42. Ferus, M., Kubelík, P., Knížek, A., Pastorek, A., Sutherland, J. & Civiš, S. (2017). High energy radical chemistry formation of HCN-rich atmospheres on early Earth. *Scientific Reports* 7, 6275. doi:10.1038/s41598-017-06489-1.
43. Choi, J. et al. (2023). Oxidation of iron by giant impact and its implication on the formation of reduced atmosphere in the early Earth. *Science Advances* 9, eadi6096. doi:10.1126/sciadv.adi6096.
44. Nesvorný, D. & Morbidelli, A. (2012). Statistical Study of the Early Solar System’s Instability with Four, Five, and Six Giant Planets. *The Astronomical Journal* 144, 117. doi:10.1088/0004-6256/144/4/117.